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A Physically Motivated Noise Model for Robust EIT Tactile Classification Under Simulated Domain Shift

A dissertation submitted to the degree of Master of Science

MSc Artificial Intelligence in Engineering and Design

Declaration of AI Use

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Abstract

Electrical Impedance Tomography (EIT) offers a promising sensing modality for electronic prosthetic skin, but classifiers trained on idealised simulation data fail on real hardware. This dissertation develops a physically motivated multi-component noise model to bridge this gap, identifies via ablation which components are necessary under deployment-realistic rather than matched evaluation, and assesses the resulting models' robustness.

A synthetic dataset of 25,000 EIT voltage measurements across five touch classes simulated a 16-electrode ionic hydrogel skin. Four classifiers were evaluated over five seeds with stratified splits, each noise component ablated under matched and deployment-realistic settings.

All models collapse to chance under domain mismatch, regardless of architecture or feature representation, but noise-matched training recovers this; the 1D-CNN reaches 75.8% $\pm$ 0.6, a 55.8-point gain. Component ablation supplies the central result: models omitting electrode positioning bias perform at chance, whereas Gaussian noise plus electrode bias alone reaches 76.5%, level with the full model. Severity scaling reveals a trade-off: fixed-noise training peaks at its training severity but loses 15.0 points by $3\times$, while randomised training is far flatter, overtaking it near $2.2\times$. Under domain shift, the models are also confidently wrong, reporting 99.7% mean confidence at 19.7% accuracy, a safety-critical failure that matched training resolves.

These findings identify the minimal noise model required for deployable EIT tactile classification, and show that the evaluation protocol conventionally used in component ablation can invert the design guidance it produces.

Acknowledgements

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List of Abbreviations

ADC Analogue-to-Digital Converter. 18, 40, 55, 59, 77

CNN Convolutional Neural Network. 35

EIT Electrical Impedance Tomography. 12–16, 18–20, 22–26, 28, 29, 31–33, 49, 52–56, 58, 59, 74

ERT Electrical Resistance Tomography. 18

FEM Finite Element Method. 19, 28

LDA Linear Discriminant Analysis. 52, 79

MLP Multi-Layer Perceptron. 15, 35, 37, 42, 81

PCA Principal Component Analysis. 35, 52, 78, 79

RF Random Forest. 15, 20, 35, 37, 42, 81

SNR Signal-to-Noise Ratio. 13, 20, 24, 25, 27, 55, 59, 74, 77

SVM Support Vector Machine. 15, 20, 35, 37, 42, 81

Chapter 1

Introduction

1.1 Background and Motivation

The sense of touch underpins safe manipulation, environmental hazard detection and social communication. Biological skin integrates four mechanoreceptor types (Purves, Augustine, Fitzpatrick, et al., 2001), providing distributed multimodal sensing across pressure, texture and temperature (Chortos, J. Liu, and Bao, 2016; Hammock et al., 2013; Shih et al., 2020). For the estimated 57.7 million people worldwide living with upper-limb loss (McDonald et al., 2021), the absence of tactile sensation in prosthetic devices is a primary driver of device abandonment, with rejection rates exceeding 35% for myoelectric prostheses (Biddiss and Chau, 2007). Restoring functional touch is therefore both a clinical priority and an enabling technology for next-generation prosthetics.

Electronic skin (e-skin) systems aim to replicate this distributed sensing in flexible, large-area substrates (Chortos, J. Liu, and Bao, 2016; Hammock et al., 2013; Shih et al., 2020). Among candidate modalities, Electrical Impedance Tomography (EIT) is attractive for prosthetics because a small number of boundary electrodes can interrogate the entire substrate, eliminating the dense internal wiring discrete sensor arrays require (H. Chen et al., 2023; Silvera-Tawil, Rye, and Velonaki, 2015). EIT reconstructs internal conductivity distributions from boundary voltage measurements (Adler and Lionheart, 2006; Cheney, Isaacson, and Newell, 1999); in ionic hydrogel substrates, applied force produces a characteristic perturbation in the measured voltage vector (Costa Cornellà et al., 2023; Park, Yuk, et al., 2022b).

Machine learning work has shown that classification of these voltage vectors can reach 98.7% accuracy on simulated tactile tasks (Park, Yuk, et al., 2022b), with comparably high figures reported elsewhere (H. Chen et al., 2023). Those accuracies are, however, obtained under idealised conditions with single-component or no noise augmentation, leaving a substantial gap between controlled demonstration and real deployment (Koo et al., 2024). Sensor morphology can itself be optimised to widen the available signal (Dong et al., 2025), but such changes also modify baseline field distributions, so hardware gains can be confounded with algorithmic ones unless morphology is held fixed, as it is here.

1.2 The Simulation-to-Reality Challenge

The deployment gap is known as the *simulation-to-reality* problem, where models trained on computationally generated data fail to maintain performance on physical hardware (Pan and Yang, 2010). EIT training data is almost exclusively produced through forward models such as EIDORS, which assume idealised electrode contact, homogeneous backgrounds and noise-free boundaries (Adler and Lionheart, 2006). Real hardware is subject to compounding degradation sources (Adler and Lionheart, 2006; Kolehmainen et al., 1997; Vilhunen et al., 2002), yet noise augmentation in prior EIT studies remains limited to single-component additive Gaussian noise (H. Chen et al., 2023; Park, Yuk, et al., 2022b). Adjacent domains have developed mature responses to analogous problems: domain randomisation (Tobin et al., 2017), adversarial domain adaptation (Ganin et al., 2016; Tzeng et al., 2017) and physically grounded corruption benchmarks (Hendrycks and Dietterich, 2019; Rusak et al., 2020), none appearing in the EIT tactile studies reviewed in Chapter 2.

1.3 Research Gap

Judged by validation practice rather than by headline accuracy, the approaches reviewed above are considerably less robust than the reported figures suggest. Park, Yuk, et al. (2022b) report 98.7% on five-class touch classification without hyperparameter tuning, repeated validation or any test under noise, so the figure reflects performance on data drawn from the same idealised distribution as training. H. Chen et al. (2023) do test under noise, showing hybrid CNN-Tikhonov reconstruction less noise-sensitive than an end-to-end network, yet the comparison holds noise at a single fixed Signal-to-Noise Ratio (SNR) of 10 dB on a single split: one operating point standing in for the range of severities a deployed sensor will meet. No system in Table 2.3 combines more than one noise source, isolates the contribution of a source once several act together, or reports whether predictive confidence remains trustworthy once the input departs from the training distribution.

The gap is defined as follows: no existing study trains or evaluates EIT tactile classification against the joint, multi-component corruption real hardware exhibits, or establishes which constituent error source is responsible for the resulting performance loss. This matters beyond benchmarking completeness. A prosthetic controller whose reported confidence does not track its correctness is unsafe to field regardless of its laboratory accuracy, and a design choice validated against a single, untested noise type has no principled basis for generalising to hardware where several distortions act at once (Section 1.2). Closing the gap therefore requires a training distribution built from the dominant physical error sources acting jointly, and an evaluation protocol able to attribute the resulting performance to each of them individually.

The first requirement concerns *training regime design*. Existing studies use single-noise augmentation or examine components in isolation, rather than treating simultaneous multi-error training as the primary benchmark. This dissertation adopts what is termed here the *foundational mode*: training on the joint distribution in which all dominant error sources act together, as they do at deployment, then asking which of those sources that mode actually requires.

The second concerns evaluation protocol, which the literature does not address. When a noise component is ablated, the resulting model may be tested either against the ablated corruption or against the full corruption a deployed device encounters. These are different questions, and Chapter 4 shows they yield opposite answers. Because real hardware exhibits all error sources simultaneously, only the latter is operationally meaningful, yet the former is the convention (Table 2.5).

1.4 Research Aims and Objectives

This dissertation investigates whether a physically motivated, multi-component noise model improves classifier robustness on noisy EIT tactile measurements, framing noise augmentation as domain randomisation (Tobin et al., 2017). The theoretical basis derives from domain adaptation theory (Ben-David et al., 2010; Pan and Yang, 2010): minimising distributional divergence between training and deployment directly reduces generalisation error. The objectives are:

1. **Develop a configurable multi-component noise model** capturing Gaussian measurement noise, contact impedance variation, electrode positioning bias and quantisation, each motivated by the experimental EIT literature (Adler and Lionheart, 2006; Kolehmainen et al., 1997; Vilhunen et al., 2002) and applied per-electrode, and **determine which components a training model must actually contain** by evaluating each ablated configuration against the full corruption encountered at deployment rather than against its own reduced corruption.

2. **Generate a physics-informed synthetic dataset** via EIDORS forward modelling (Adler and Lionheart, 2006), simulating a 16-electrode ionic hydrogel skin across five touch classes.

3. **Compare classical and deep classifiers** under identical noise conditions, benchmarking Support Vector Machine (SVM), Random Forest (RF) and Multi-Layer Perceptron (MLP) baselines against a 1D-CNN to establish whether noise robustness is determined by model capacity or by architecture.

4. **Evaluate robustness** across noise severities (Hendrycks and Dietterich, 2019) and across training regimes (fixed, randomised, mixed), establishing how regime choice depends on how well the deployment environment is characterised.

1.5 Contributions

The work's novel contributions are: a **structured per-electrode noise model**, applying interface effects per-electrode rather than per-measurement to reproduce the block-structured distortion real hardware exhibits; a **minimal sufficient noise model**, establishing which components are necessary and redundant; a **correction to ablation evaluation protocol**, showing that evaluating ablated models against deployment rather than ablated corruption reverses which components appear to matter, a distinction none of the reviewed studies draws and one applicable beyond EIT; and an **open, reproducible pipeline** addressing the standardisation gap identified by Koo et al. (2024).

1.6 Scope

The study is bounded so that any robustness improvement is attributable to the training strategy rather than confounded by hardware or temporal variability. All data are generated by forward modelling, since controlling test-time corruption answers the research question without conflating noise-model accuracy with classifier robustness. Classification is static and single-frame, matching the dominant EIT formulation (H. Chen et al., 2023; Park, Yuk, et al., 2022b) and excluding temporal effects such as drift. The forward model is a 2D cross-section, necessary for 25,000 solves per mesh; sensor morphology is held fixed (Dong et al., 2025); and the five touch classes represent archetypal prosthetic interactions (Silvera-Tawil, Rye, and Velonaki, 2012) without exhausting the space of tactile events. The consequences are examined in Section 5.7.

Chapter 2

Literature Review

2.1 Electronic Skin for Prosthetic Applications

2.1.1 Biological Motivation and Clinical Need

The clinical demand motivating this work (Section 1.1) has driven extensive research into e-skin platforms that replicate biological skin's compliance, multimodal sensing and drift robustness within a manufacturable substrate (Chortos, J. Liu, and Bao, 2016; Hammock et al., 2013; Shih et al., 2020). Choo and Chang (2023) reviews machine learning in prosthetics and orthotics, finding high accuracy on narrow clinical tasks but inconsistent reporting that prevents cross-study comparison and leaves generalisation across users unestablished.

2.1.2 Material Platforms for Electronic Skin

The choice of substrate constrains sensing performance, durability and wearability. Table 2.1 evaluates the dominant platforms against four criteria bearing on prosthetic deployment.

Hydrogels (illustrated in Figure 2.1) approximate biological tissue most closely in compliance and biocompatibility (Park, Yuk, et al., 2022b; Tao et al., 2023), and ionic hydrogels exhibit a *positive piezoresistive response*: compression disrupts ion transport, raising local resistance under load (Costa Cornellà et al., 2023). Silicone encapsulation improves sensitivity but is tested only to $\sim$100 cycles (Dong et al., 2025). **Conductive elastomers** remove moisture dependence at the cost of non-linearity and fatigue-driven degradation (Kim et al., 2024), confirming that no platform eliminates drift. **Printed arrays** (Yu et al., 2024) are scalable but limited to 81 discrete taxels, incompatible with distributed EIT sensing.

The central insight is that while each platform introduces *different* dominant material noises, the measurement instrumentation contributes a *shared* set of per-frame

Table 2.1: Comparative assessment of electronic skin material platforms for prosthetic applications.

Platform	Stability	Durability	Fidelity	Scalability
Hydrogels	Poor	Moderate	Excellent	Moderate
Hydrogel + silicone lattice encapsulation	Moderate	Poor	Excellent	Moderate
Conductive elastomers	Good	Excellent	Good	Good
Printed arrays	Good	Moderate	High sensitivity but low resolution	Excellent

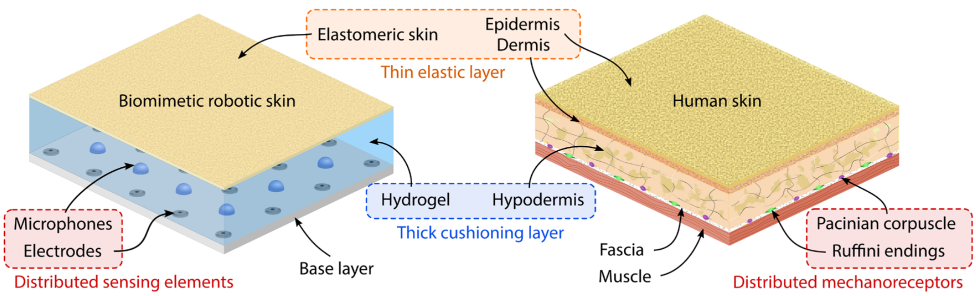


Figure 2.1: Biomimetic hydrogel-elastomer robotic skin compared with human skin. Both combine a thin elastic outer layer (elastomeric skin; epidermis/dermis) with a thick, compliant cushioning layer (hydrogel; hypodermis) housing distributed sensing elements or mechanoreceptors, respectively, motivating the choice of substrate detailed in Section 3.1.2. Reproduced from Park, Yuk, et al. (2022b).

distortions regardless of substrate: Analogue-to-Digital Converter (ADC) quantisation, electrode–substrate contact impedance, positioning uncertainty and thermal noise. This two-level taxonomy justifies targeting the shared measurement-chain distortions, giving cross-platform generalisability without substrate-specific retraining.

2.1.3 A Comparative View of Sensing Modalities

The choice of modality determines the nature of the sim-to-real challenge. Piezoresistive and capacitive arrays suffer discrete failures localised to the affected taxel, whereas in EIT every boundary measurement is a function of the conductivity field as a whole, so a disturbance at one electrode perturbs the entire vector. Corruption is therefore distributed and structured rather than sparse and local, which motivates a modality-specific noise model. EIT also needs only current injection and voltage measurement, unlike the costly interrogators fibre Bragg gratings require (Massari et al., 2022; Silvera-Tawil, Rye, and Velonaki, 2015). Silvera-Tawil, Rye, and Velonaki (2012) demonstrated one of the earliest EIT touch classifiers, relying on handcrafted representations.

Table 2.2: Comparison of sensing modalities for electronic skin.

Modality	**Advantages**	**Limitations**	**Representative Work**
Piezoresistive arrays	High sensitivity, simple readout	Dense wiring, fragile interconnects	Yu et al. (2024)
Capacitive arrays	Good linearity, low hysteresis	Cross-talk, humidity sensitivity	Hammock et al. (2013)
Optical (FBG)	Distributed sensing, EMI immune	Bulky interrogators, cost	Massari et al. (2022)
EIT/ERT	Few electrodes, large area, flexible	Ill-posed reconstruction, low resolution	H. Chen et al. (2023) and Wang et al. (2025)
Acoustic tomography	Complementary to EIT	Complex hardware, noise	Park, Yuk, et al. (2022b)

2.2 Electrical Impedance Tomography for Tactile Sensing

2.2.1 Principles of EIT

EIT reconstructs the internal conductivity distribution $\sigma(\mathbf{x})$ of a body Ω from boundary voltage measurements: small alternating currents are injected through electrode pairs at the boundary $\partial\Omega$, and the resulting voltage differences are measured across the remaining pairs, governed by the generalised Laplace equation:

$$\nabla \cdot (\sigma(\mathbf{x})\nabla u(\mathbf{x})) = 0, \quad \mathbf{x} \in \Omega \tag{2.1}$$

where $u(\mathbf{x})$ is the electric potential. The forward problem is well-posed and solved using Finite Element Method (FEM) (Adler and Lionheart, 2006; Cheney, Isaacson, and Newell, 1999), but recovering σ from boundary measurements is ill-posed, so small measurement errors can produce large, non-physical artefacts (Cheney, Isaacson, and Newell, 1999). Dimas et al. (2024) survey the resulting shift in EIT solvers from regularisation to deep learning.

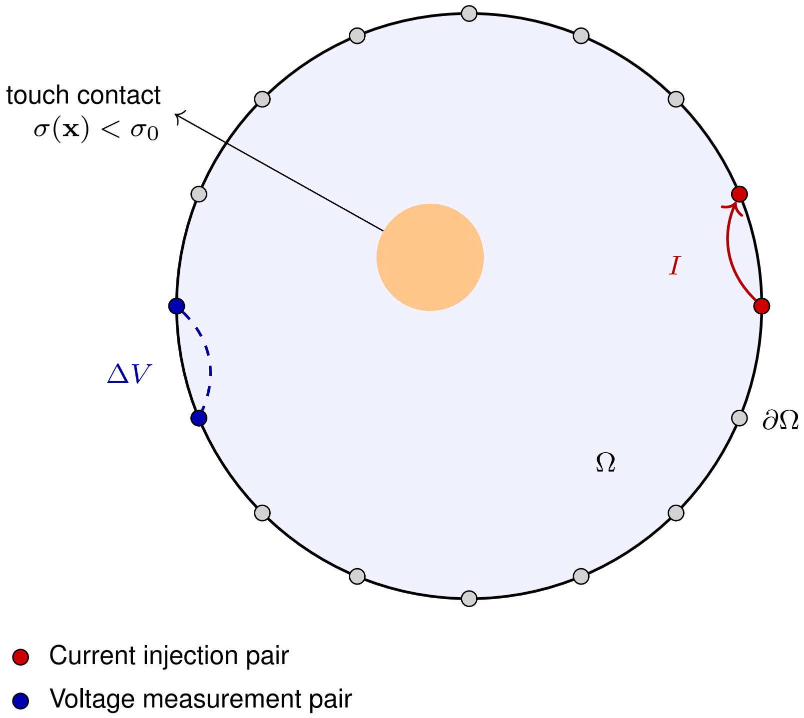


Figure 2.2: Schematic of the EIT forward problem for a 16-electrode boundary array. Alternating current I is injected through an adjacent electrode pair (red); the resulting voltage difference ΔV is measured across another electrode pair (blue) elsewhere on the boundary $\partial\Omega$. A local conductivity perturbation $\sigma(\mathbf{x}) < \sigma_0$ (orange), such as a touch contact, alters the measured voltage vector relative to the unperturbed background. The adjacent drive pattern (Section 2.2.2) repeats this injection sequentially around all 16 electrodes.

2.2.2 EIT Measurement Patterns and Data Representation

For n_e electrodes, the adjacent drive pattern injects current through neighbouring pairs and measures across the remainder, yielding $n_e(n_e - 3)$ measurements per frame, or $16 \times 13 = 208$ for a 16-electrode system (Adler, Arnold, et al., 2009) (Figure 2.2). Park, Lee, et al. (2022a) proposes adaptive acquisition of the most informative patterns to reduce time and energy, though its selection criterion assumes a known noise model, precisely what this work shows is usually mis-specified.

2.2.3 EIDORS

EIDORS provides standardised forward models, inverse solvers and meshing tools for EIT, though Adler and Lionheart (2006) cautions that its flexibility invites misuse. Those cautions apply equally to ML models trained on EIDORS data, which inherit the forward model's assumptions of homogeneous background, idealised electrode contact and noise-free boundaries. Mitigation is detailed in Section 3.1.1 and evaluated in Section 5.4.

2.2.4 EIT-Based Electronic Skin Systems

Table 2.3 summarises recent EIT-based tactile sensing systems for robotic and prosthetic applications.

H. Chen et al. (2023) propose CNN-enhanced Tikhonov regularisation for EIT pressure reconstruction, showing pure end-to-end ML is more noise-sensitive than hybrid methods, but train only with additive Gaussian noise at SNR = 10 dB on a single split. Dong et al. (2025) show sensor morphology alters EIT observability before any ML optimisation, a lattice architecture improving 12-class classification to a reported 99.6%, though morphology gains are not commensurable with noise-aware training gains on a fixed sensor, and their augmentation remains Gaussian-only.

Costa Cornellà et al. (2023) characterise an ionic–electronic dual-conductivity hydrogel in a 16-electrode configuration with a feedforward network for localisation, establishing that ionic hydrogels produce measurable conductivity *decreases* under load, though no gauge-factor calibration is provided and evaluation uses only 500 samples on a single 80/10/10 split, leaving the force-to-conductivity mapping and noise robustness uncharacterised. Kim et al. (2024) observe gradual degradation over 10,000 cycles and identify temperature as a key noise source; their ultrathin elastomer ($E \sim$150 kPa) informs the Hertzian contact geometry used in Chapter 3.

2.3 Machine Learning for Tactile Classification

2.3.1 Classical Machine Learning Approaches

Classical methods dominated tactile classification before deep learning. SVMs (Cortes and Vapnik, 1995) suit high-dimensional data with few samples, Rahiminejad et al. (2021) reporting near-perfect accuracy on spike-encoded features. RFs (Breiman, 2001)

Table 2.3: Summary of EIT-based electronic skin systems in the literature.

Study	**Electrodes**	**Material**	**ML Method**	**Key Limitation**
Park, Yuk, et al. (2022b)	16	Hydrogel–elastomer	CNN (5 classes)	No cross-validation
H. Chen et al. (2023)	16	Carbon-black composite	CNN-TR hybrid	Single noise type
Wang et al. (2025)	Variable	Granular jamming	Reconstruction only	Calibration per stiffness
Costa Cornellà et al. (2023)	16	Ionic hydrogel	Feedforward NN	No gauge-factor calibration
Kim et al. (2024)	16	Ultrathin elastomer	Neural network	Long-term drift
Jiang et al. (2024)	16	Hydrogel	FISTA algorithm	Conductive objects only

add feature-importance ranking and scale invariance. Costa Cornellà et al. (2023) used a feedforward network for EIT localisation on 500 samples with a single split, so whether such compact architectures survive real measurement distortions is unknown.

Chapter 3 adopts all three as baselines, using hyperparameters fixed *a priori* from canonical practice rather than the values reported for these different datasets and tasks (Section 3.5.2).

2.3.2 Convolutional Neural Networks for Tactile Data

CNNs dominate tactile classification through their capacity to learn hierarchical features without manual engineering (LeCun, Bengio, and Hinton, 2015), suiting the nonlinear voltage-to-contact mapping. Park, Yuk, et al. (2022b) report 98.7% on 5-class touch classification, but without hyperparameter optimisation, repeated validation or noise testing. For 1D voltage-difference vectors, a 1D-CNN is architecturally natural, with kernels learning local patterns across adjacent electrode pairs.

2.3.3 Deep Learning for EIT Reconstruction

Deep learning has also been applied to EIT reconstruction itself (Hamilton and Hauptmann, 2018; Wei and X. Chen, 2020), though validated primarily on simulation with idealised noise. Cebrian Carcavilla and Meribout (2025) distinguish data-driven approaches, learning direct voltage-to-image mappings, from model-based methods incorporating forward-model constraints, arguing the former generalise poorly and lack physical consistency guarantees, favouring physics-informed approaches for safety-critical use.

The distinction between reconstruction and classification matters: this work adopts classification, bypassing reconstruction entirely, since a prosthetic controller requires a touch *category* rather than a spatial map, asking whether noise-aware training can deliver robustness comparable to the hybrid physics-plus-ML approaches of H. Chen et al. (2023) without explicit physics-based reconstruction.

2.3.4 Methodological Concerns Across the Literature

Several concerns recur *systematically*, catalogued in Table 2.5: reported accuracies rest predominantly on single train/test splits (H. Chen et al., 2023; Park, Yuk, et al., 2022b; Tao et al., 2023; Yu et al., 2024), training data come from single phantoms, and design choices are seldom isolated by ablation. None of the studies reviewed evaluates against a standardised corruption benchmark comparable to ImageNet-C (Hendrycks and Dietterich, 2019), and Goodfellow, Shlens, and Szegedy (2015) established that perturbation vulnerability is intrinsic to high-dimensional models.

2.4 The Simulation-to-Reality Gap

2.4.1 Origin of the Gap in EIT Systems

The simulation-to-reality gap arises when models trained on computationally generated data fail on physical hardware. Ben-David et al. (2010) formalise this: target-domain error is bounded by source-domain error plus distributional divergence, so reducing that divergence directly improves generalisation. In EIT, this divergence has several compounding origins.

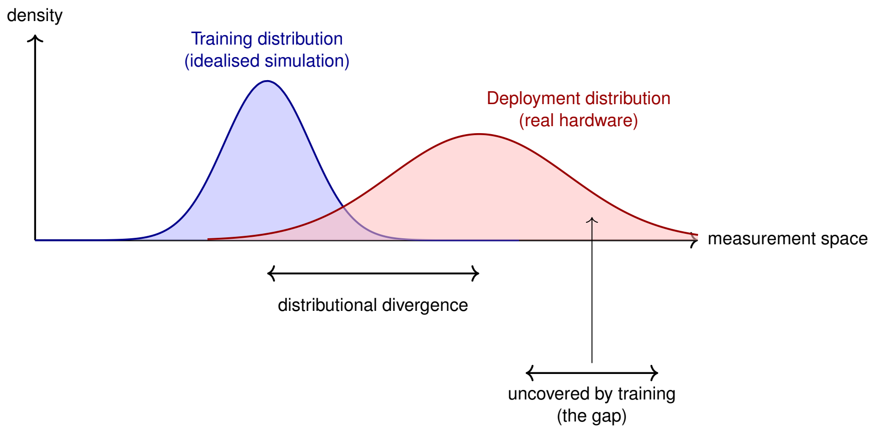


Figure 2.3: Schematic of the simulation-to-reality gap as a distributional divergence (Ben-David et al., 2010). The training distribution (blue), produced by an idealised forward model, and the deployment distribution (red), produced by real hardware, overlap only partially. The region under the deployment curve lying outside the training distribution's support (right) receives no representation during training and is where a classifier trained only on clean simulation is expected to fail.

Measurement noise is statistically richer than the additive Gaussian model usually assumed: electrode–skin contact impedance imposes multiplicative, spatially varying distortion (Vilhunen et al., 2002), positioning errors impose systematic bias (Kolehmainen et al., 1997), and skin-impedance change drives temporal drift (Boone and Holder, 1996). *Material non-idealities* compound this: moisture-dependent conductivity (Park, Yuk, et al., 2022b; Tao et al., 2023) and temperature sensitivity (Kim et al., 2024). *Model simplifications* such as 2D approximation and idealised electrodes (Adler and Lionheart, 2006) add systematic bias random augmentation cannot compensate (Section 5.4).

Which sources a noise model can capture. Table 2.4 states which of these fall within the scope of a per-frame static noise model: sources stationary within a frame can be represented as a stochastic perturbation of the measurement vector, whereas those defined by change *across* frames, or by domain deformation, cannot. Shih et al. (2020) note the growing use of recurrent architectures for sequential contact dynamics, but no reviewed study applies them to compensate for progressive drift.

The sources the field has invested most effort in compensating for, electrode positioning and contact impedance, for which dedicated estimation procedures exist (Kolehmainen et al., 1997; Vilhunen et al., 2002), are precisely those within scope: a static model should therefore capture a substantial share of the gap despite its incompleteness. That these long-documented errors are seldom represented in modern ML-based EIT training distributions is the gap this dissertation addresses.

Table 2.4: Documented EIT error sources and their representability within a per-frame static noise model.

Source	In scope	Rationale
Thermal and amplifier noise	Yes	Additive noise in the analogue chain; modelled at fixed SNR (Adler and Lionheart, 2006)
Contact impedance variation	Yes	Must be modelled or estimated; motivates the Complete Electrode Model (Somersalo, Cheney, and Isaacson, 1992; Vilhunen et al., 2002)
Electrode positioning error	Yes	A dominant modelling error in static imaging (Kolehmainen et al., 1997)
ADC quantisation	Yes	Standard uniform quantiser model (Widrow, Kollár, and M.-C. Liu, 1996)
Temporal / baseline drift	No	Sequential across frames; undefined within a single static measurement (Boone and Holder, 1996)
Material fatigue, temperature	No	Evolves over thousands of cycles (Kim et al., 2024)
Boundary shape deformation	No	Alters domain geometry and the sensitivity map, not merely the measurement (Kolehmainen et al., 1997)
Electrode size mismatch	No	Degrades reconstruction as mismatch grows; fixed electrode geometry assumed here (Kolehmainen et al., 1997)
Frequency-dependent impedance	No	Requires multi-frequency acquisition; single-frequency assumed here
Motion artefacts, mains pickup	No	Non-stationary or environment-specific

2.4.2 Domain Adaptation and Randomisation

Tobin et al. (2017) introduced domain randomisation: with sufficient simulated variability, the real distribution becomes a subset of the training distribution. Ganin et al. (2016) and Tzeng et al. (2017) developed domain-adversarial alternatives, resembling instance-based transfer in the taxonomy of Pan and Yang (2010); neither appears in the EIT tactile studies of Table 2.3, and both need paired data that this study lacks.

Hendrycks and Dietterich (2019) introduced ImageNet-C, showing state-of-the-art classifiers suffer catastrophic drops under corruptions humans handle trivially, and established the multi-severity evaluation paradigm adopted in Chapter 3. Rusak et al. (2020) showed simple Gaussian augmentation gives limited corruption robustness against diverse physically grounded strategies.

2.4.3 Noise Augmentation as a Transfer Strategy

Whether existing EIT noise models suffice for domain randomisation is doubtful: H. Chen et al. (2023) add white Gaussian noise at fixed SNR and Park, Yuk, et al. (2022b) augment with Gaussian noise and background samples, neither considering contact impedance, electrode bias or quantisation. GhasemAzar and Vahdat (2007) showed networks trained on noisy EIT data perform better under noise but worse on clean data. No reviewed study asks *which* components matter, important since Geirhos et al. (2018) found networks generalise poorly across distortion types.

2.4.4 Limitations and Counter-Arguments

Three conditions under which noise augmentation may fail warrant attention: the **precision–robustness trade-off** (GhasemAzar and Vahdat, 2007) may cost accuracy without benefit under well-controlled, frequently recalibrated deployment, so this work evaluates across a *range* of severities; **model mismatch** yields only false robustness from an omitted or misspecified dominant distortion (Geirhos et al., 2018); and **alternative approaches** such as domain adaptation (Ganin et al., 2016; Tzeng et al., 2017) may be superior where noise resists parametric modelling.

2.5 Deployment Considerations

Prosthetic systems operate within tight power, weight and computational budgets, and Jiang et al. (2024) argue that classical algorithms suit embedded deployment better than deep learning. Sensor density compounds this: greater density means more wiring and reduced flexibility (Koo et al., 2024; Shih et al., 2020), which is why EIT senses distributively from few boundary electrodes (Silvera-Tawil, Rye, and Velonaki, 2015), at the cost of an ill-posed reconstruction. Ethical and economic implications, including the risk that high-cost sensing widens healthcare inequality (McDonald et al., 2021), are addressed in Section 5.6.

2.6 Summary and Identification of Research Gaps

The field has advanced strongly in materials, sensing hardware and ML architectures, but significant gaps remain in validation methodology and deployment readiness (Table 2.5), demonstrating the *feasibility* of ML-enabled EIT tactile sensing without establishing its *reliability*. As introduced in Section 1.3, the gap is defined as the absence of both a training distribution built from the dominant physical error sources acting jointly, as real hardware presents them, and an evaluation protocol able to attribute the resulting performance to each source individually; Table 2.5 shows that no reviewed system provides either. The methods conventionally used to approximate the latter, evaluating an ablated component against its own reduced corruption rather than the full deployment corruption, can point in the wrong direction (Section 4.2). This dissertation quantifies that distance, decomposing it by noise source and verifying sensitivity to the modelling assumptions.

Table 2.5: Summary of identified research gaps and how the present work addresses them.

Gap Identified	**Evidence**	**Addressed in This Work**
Single-component noise models only	H. Chen et al. (2023) and Park, Yuk, et al. (2022b)	Four-component physically motivated noise model applied as domain randomisation (Section 3.2)
No ablation of noise contributions; where components are isolated, models are tested only against the reduced corruption	None of the studies in Table 2.3 reports an ablation	Per-component ablation evaluated against the *full* deployment corruption (Section 4.2)
Noise used only as training augmentation, never as an evaluation axis	H. Chen et al. (2023) at a single SNR, Dong et al. (2025) across 35–65 dB, Park, Yuk, et al. (2022b) as a fivefold augmentation	Multi-severity sweep across three training regimes (cf. ImageNet-C (Hendrycks and Dietterich, 2019))
Predictive confidence not reported	Absent from every study in Table 2.3	Confidence compared against achieved accuracy (Section 4.4.2)
Missing standardised benchmark	Koo et al. (2024)	Open-source dataset and reproducible framework

Chapter 3

Research Methods

This chapter details the simulation-based design summarised in Figure 3.1. All code, configuration files and trained models are publicly available, addressing the standardisation gap of Chapter 2; the repository URL is in Appendix C.2.

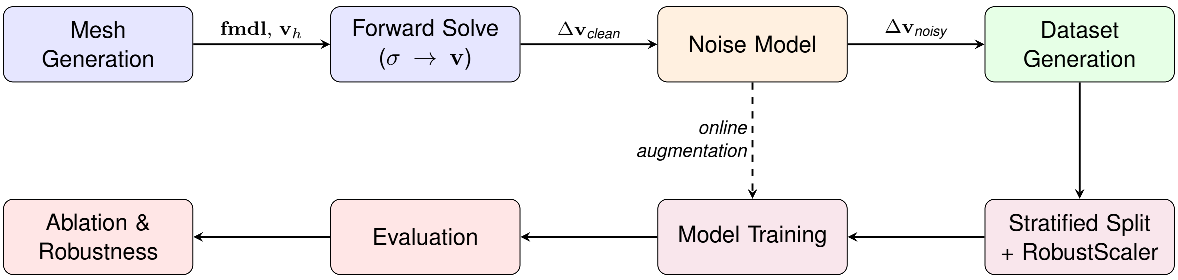


Figure 3.1: End-to-end methodology pipeline. The upper row is MATLAB/EIDORS data generation, read left to right; the lower row is the Python/PyTorch training and evaluation path, read right to left. Solid arrows carry the data artefacts named beside them. The dashed arrow marks the same noise model being re-applied at training time for the online-augmentation and ablation regimes, so noise enters the pipeline at two distinct points rather than one.

3.1 EIT Simulation Framework

3.1.1 Forward Model and Mesh Configuration

The EIT forward problem is solved using EIDORS v3.12-ng (Adler and Lionheart, 2006), computing boundary voltages from a specified conductivity distribution via the FEM. The governing equation is the generalised Laplace equation (Equation 2.1 in Chapter 2), solved with Neumann boundary conditions and the Complete Electrode Model (CEM) to represent finite electrode area and contact impedance (Cheney, Isaacson, and Newell, 1999; Somersalo, Cheney, and Isaacson, 1992).

The domain is a 2D circular cross-section representing a transverse slice of a prosthetic forearm sensor, discretised using `mk_common_model('d2C', 16)`. This canonical

EIDORS reference family was preferred over a custom mesh so that methodological contributions are not confounded with bespoke meshing choices (Adler, Arnold, et al., 2009; Adler and Lionheart, 2006), and 2D was retained for tractability at dataset scale (25,000 forward solves per mesh) and consistency with prior EIT classification work.

Mesh selection was evidence-based rather than assumed: three matched noisy datasets were generated at coarse, medium and fine refinement and a CNN trained per mesh was cross-evaluated on all three. Mesh d (medium) was retained, transferring more stably across meshes (4.37 pp cross-mesh range) than the coarse alternative (8.35 pp) at comparable runtime, so the choice reflects generalisation rather than cost. Full cross-mesh figures and compute profiles are in Appendix A.1.

3.1.2 Material Model: Ionic Hydrogel

The simulated substrate models an ionic hydrogel layer, following the material system demonstrated by Park, Yuk, et al. (2022b) and characterised by Costa Cornellà et al. (2023), exhibiting the positive piezoresistive response described in Section 2.1.2 and the multilayer structure of Figure 3.2.

Hydrogel was preferred to the more durable conductive elastomers of Table 2.1 for two reasons: it is the substrate used by the most directly comparable EIT e-skin studies (Costa Cornellà et al., 2023; Jiang et al., 2024; Park, Yuk, et al., 2022b), and Costa Cornellà et al. (2023) provides a piezoresistive characterisation in an EIT configuration that no equivalent elastomer study offers.

The forward model assigns conductivity $\sigma < \sigma_0$ within the contact region, decreasing with applied pressure. Background conductivity is normalised to $\sigma_0 = 1.0$ S/m, standard in EIDORS-based studies (Adler and Lionheart, 2006); this does not affect the voltage *difference* vectors used for classification, which depend only on the contrast ratio σ/σ_0.

3.1.3 Touch Class Definitions

Five touch classes represent the principal interaction modalities encountered in prosthetic use (Park, Yuk, et al., 2022b; Silvera-Tawil, Rye, and Velonaki, 2012), each parameterised by a conductivity range (σ) and a spatial radius (r), with random position constrained to the inner 40% of the mesh (Table 3.1).

Three design principles govern this parameterisation:

1. **Sign constraint.** All contact classes satisfy $\sigma < \sigma_0$, consistent with ionic hydrogels' positive piezoresistive response (Costa Cornellà et al., 2023).

2. **Pressure-driven conductivity change.** The conductivity drop is proportional to local pressure $P = F/(\pi r^2)$ rather than force alone, which separates the classes

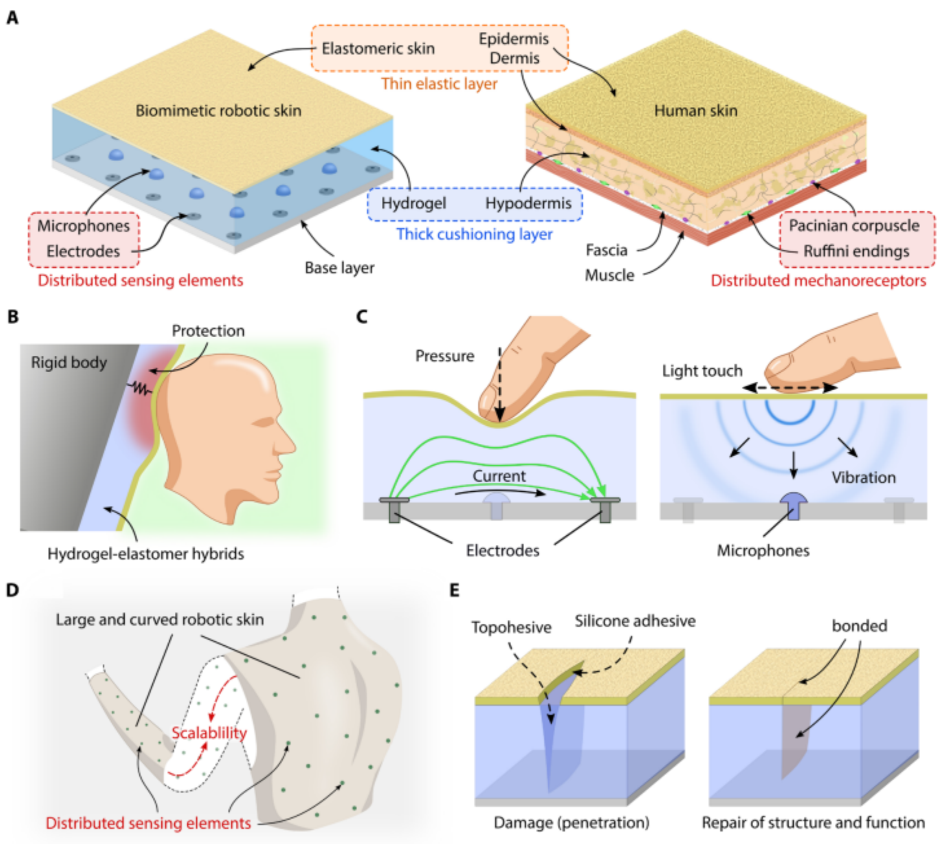


Figure 3.2: Concept of biomimetic multilayer structure for robotic skin. (A) Comparison between the biomimetic multilayer structure based on hydrogel-elastomer hybrids and human skin. Both structures consist of an elastic, tough outermost layer (elastomeric skin and dermal layer) and a thick, compliant cushioning layer (hydrogel and hypodermis). Sensing elements were also evenly distributed deep in the multilayer structure. (B) The multilayer structure allows the robotic skin to achieve appropriate material properties and dimensions for the protection function. (C) Because the hydrogel layer is compliant and electrically conductive, it can be used as a pressure-sensitive material. In addition, dynamic tactile stimuli (light touch) can be measured by detecting vibrations generated on the surface and transmitted through the multilayer structure. (D) Scalable design based on distributed sensing elements with a large and overlapping receptive field. (E) The biomimetic robotic skin can be repaired with an adhesive after damage to restore its structure and function. Reproduced from Park, Yuk, et al. (2022b).

Table 3.1: Touch class parameterisation. Conductivity and radius ranges are sampled uniformly for each generated sample. $\Delta\sigma$ denotes the percentage decrease from the homogeneous baseline.

Class	σ **(S/m)**	$\Delta\sigma$	**Radius**
No contact	1.00	0%	—
Light touch	[0.85, 0.95]	5–15%	[0.06, 0.10]
Firm press	[0.55, 0.75]	25–45%	[0.08, 0.12]
Point contact	[0.35, 0.55]	45–65%	[0.02, 0.05]
Distributed	[0.80, 0.92]	8–20%	[0.15, 0.25]

physically: point contact (small area, high pressure) produces the largest per-element $\Delta\sigma$ and distributed contact (large area, low pressure) the smallest. Point and distributed contacts therefore have comparable signal energy but different spatial distributions.

3. **Contact geometry.** Radii follow Hertzian contact mechanics for soft elastomers: for $E^* \approx 150$ kPa (Kim et al., 2024) and forces of 0.1–4 N, predicted radii span approximately 2–25 mm on the unit-normalised mesh (Table 3.1).

Position constraint. Contact positions are restricted to $|x|, |y| \leq 0.4$ on the unit disc, since EIT sensitivity is far greater near the boundary electrodes than at the centre (Adler and Lionheart, 2006), and the prosthetic's sensing surface is the top of the sensor, not the electrode boundary. An earlier, wider constraint ($|x|, |y| \leq 0.7$) let within-class positional variation dominate over between-class differences, motivating the tighter bound.

Conductivity magnitudes are *physics-informed priors* rather than measured calibration constants, since no published study provides a complete gauge-factor calibration for the ionic hydrogel under these conditions (Costa Cornellà et al., 2023).

Each sample is the voltage difference vector $\Delta\mathbf{v} = \mathbf{v}_{\text{contact}} - \mathbf{v}_{\text{baseline}} \in \mathbb{R}^{208}$ introduced in Section 2.2.2, where $\mathbf{v}_{\text{baseline}}$ is the homogeneous no-touch reference. Differencing removes the large common-mode baseline and mirrors real EIT operation, which acquires a reference frame during calibration (Adler and Lionheart, 2006).

3.2 Noise Model

The noise model comprises four independently configurable components, each capturing a distinct physical degradation source documented in the experimental EIT literature.

3.2.1 Design Rationale

Real EIT hardware corrupts measurements through multiple concurrent, statistically distinct processes (Adler and Lionheart, 2006; Boone and Holder, 1996; Vilhunen et al., 2002), whereas prior work models at most one, invariably Gaussian noise (H. Chen et al., 2023; Park, Yuk, et al., 2022b). Training with all four dominant per-frame distortions active simultaneously is termed the *foundational mode*: the classifier is exposed to the co-occurring errors expected at deployment.

Temporal drift is deliberately excluded, being a *sequential* phenomenon that cannot be represented within a single static frame; modelling it as a spatial random walk across channels would introduce artefacts with no hardware counterpart. Addressing it requires sequential data (Chapter 6).

3.2.2 Component Specifications

All components operate on the voltage difference vector $\Delta\mathbf{v} \in \mathbb{R}^{208}$ and are applied sequentially in a fixed order (Gaussian $\rightarrow$ contact impedance $\rightarrow$ bias $\rightarrow$ quantisation) reflecting the physical signal chain: measurement noise enters at the analogue front-end, followed by electrode-level effects, geometric bias, and finally digitisation. The contact-impedance and electrode-bias components are applied *per-electrode* rather than per-measurement, since these are properties of the interface. Each of the 16 electrodes carries a single value mapped to blocks of the measurement vector, producing the block-structured distortion real hardware exhibits. Unless stated otherwise, every parameter magnitude below is a nominal design point rather than a measured calibration constant, bounded by the explicit sensitivity sweeps of Section 3.6.6 and summarised in Table 3.2.

Component 1: Gaussian measurement noise. Models the aggregate effect of thermal noise, amplifier noise, and electromagnetic interference in the analogue measurement chain (Adler and Lionheart, 2006; Boone and Holder, 1996).

$$\Delta\mathbf{v}' = \Delta\mathbf{v} + \mathbf{n}, \quad \mathbf{n} \sim \mathcal{N}(\mathbf{0},\, \sigma_n^2\mathbf{I}), \quad \sigma_n = \begin{cases} \frac{\|\Delta\mathbf{v}\|_2}{\|\hat{\mathbf{n}}\|_2} \cdot 10^{-\mathsf{SNR}/20} & \text{if } \|\Delta\mathbf{v}\| > 0 \\ \frac{n_{\mathsf{floor}} \cdot N}{\|\hat{\mathbf{n}}\|_2} & \text{if } \|\Delta\mathbf{v}\| = 0 \end{cases} \tag{3.1}$$

where $\hat{\mathbf{n}} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and $N = 208$ (SNR and noise-floor values in Table 3.2). No agreed instrumentation figure exists; published EIT studies add noise anywhere between 10 and 65 dB (H. Chen et al., 2023; Dong et al., 2025), so 40 dB is taken as a mid-range value. The noise floor is essential, as without it, the no-contact class remains an exact zero vector, a deterministic cluster with no hardware counterpart.

Component 2: Electrode contact impedance variation. Models imperfect and spatially varying electrode–substrate contact, producing multiplicative per-electrode distortion.

$$\Delta v_i' = \Delta v_i \cdot z_{e(i)}, \quad z_e = \exp(\epsilon_e), \quad \epsilon_e \sim \mathcal{N}(0,\, \sigma_z^2), \quad \sigma_z = 0.10 \tag{3.2}$$

where $e(i)$ maps measurement i to its electrode block ($e \in \{1, \ldots, 16\}$), so each electrode contributes a single factor to all measurements it participates in, producing block-structured multiplicative distortion. The log-normal form guarantees $z_e > 0$ and yields asymmetric variation about unity. Vilhunen et al. (2002) show that contact impedance cannot be neglected in static imaging and must be estimated jointly with the interior admittivity, motivating its explicit inclusion here.

This default is the least well-constrained parameter in the model: no published study provides electrode–hydrogel contact-impedance *variation* data suitable for direct parameterisation (Costa Cornellà et al., 2023; Park, Yuk, et al., 2022b), though the physical analogy of a metal electrode contacting an ionic conductor supports the same order of magnitude.

Component 3: Electrode positioning bias. Models systematic measurement error from imprecise electrode placement on the physical substrate.

$$\Delta v_i' = \Delta v_i + b_{e(i)}, \quad b_e \sim \mathcal{U}(-b_{\max}, b_{\max}), \quad b_{\max} = 0.02 \tag{3.3}$$

where each electrode receives an independently drawn fixed offset, reflecting that placement error at one site is unrelated to the others, and producing block-structured additive bias. Kolehmainen et al. (1997) identify electrode placement as a dominant modelling error in static imaging, and an independent per-electrode draw captures its block structure more faithfully than a linear gradient across measurements. The independence assumption is revisited as a limitation in Section 5.7, since this component proves to dominate the domain gap.

Component 4: Quantisation noise. Models the finite precision of the analogue-to-digital converter (ADC) as additive uniform noise bounded by half the least-significant bit, the standard statistical model for a mid-tread uniform quantiser operating well above its noise floor (Widrow, Kollár, and M.-C. Liu, 1996).

$$\Delta v_i' = \Delta v_i + q_i, \quad q_i \sim \mathcal{U}\left(-\frac{\text{LSB}}{2}, \frac{\text{LSB}}{2}\right), \quad \text{LSB} = \frac{V_{\text{range}}}{2^{n_{\text{bits}}}} \tag{3.4}$$

with $n_{\text{bits}} = 16$ and $V_{\text{range}} = 1.0\,\text{V}$ ($\text{LSB} \approx 1.5 \times 10^{-5}\,\text{V}$), corresponding to a typical EIT data-acquisition specification and setting the minimum noise floor of the digital chain.

3.2.3 Parameter Summary and Ablation Design

Table 3.2 consolidates all noise parameters with their physical sources.

3.3 Dataset Generation

3.3.1 Sampling Procedure

The dataset comprises 25,000 samples (5,000 per class), generated using the MATLAB pipeline with a fixed seed ($\texttt{seed} = 42$) for reproducibility. For each sample:

1. Select the target class and sample conductivity $\sigma \sim \mathcal{U}[\sigma_{\min}, \sigma_{\max}]$, radius $r \sim \mathcal{U}[r_{\min}, r_{\max}]$, and position $(x, y) \sim \mathcal{U}[-0.4, 0.4]^2$.

Table 3.2: Noise model parameters. The cited works establish that each mechanism matters; the magnitudes are nominal and bounded by the sweeps of Section 3.6.6.

Component	Parameter	Value	Source
Gaussian	SNR	40 dB	Nominal; swept 20–60 dB (Section 3.6.6)
Gaussian	Noise floor	10^{-4} V	Typical EIT baseline noise
Contact impedance	σ_z	10%	Nominal; swept 5–25% (Section 3.6.6)
Electrode bias	$b_{\max}$	0.02	Nominal; swept 0.005–0.08 (Section 3.6.6)
Quantisation	bits / V_{range}	16 / 1.0 V	Typical EIT ADC specifications

2. Construct the inhomogeneous conductivity image: $\sigma(\mathbf{x}) = \sigma$ for elements within radius r of (x, y); σ_0 elsewhere.
3. Solve the forward problem to obtain $\mathbf{v}_{\text{contact}}$.
4. Compute $\Delta\mathbf{v}_{\text{clean}} = \mathbf{v}_{\text{contact}} - \mathbf{v}_{\text{baseline}}$.
5. Apply the full noise model to produce $\Delta\mathbf{v}_{\text{noisy}}$.
6. Store both clean and noisy vectors with the class label and generation metadata.

Storing both vectors enables the four-condition evaluation matrix (Section 3.6.1) without regenerating data. Dataset size was set empirically: a size sweep (Figure 3.3) shows noise-aware accuracy plateauing near 10^4 samples, so 25,000 sits beyond the plateau within a practical generation budget, while the clean-trained curve remains flat at chance throughout, an early sign that no quantity of clean data addresses the domain shift.

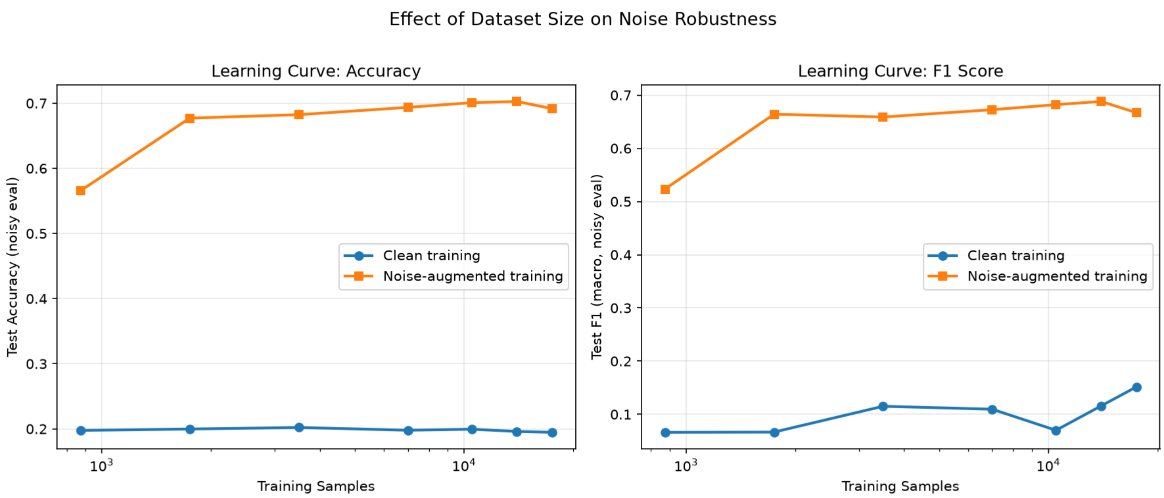


Figure 3.3: Effect of dataset size on test accuracy (left) and F1 Score (right), evaluated on noisy data. The noise-aware model improves steeply, then plateaus near 10^4 samples; the clean-trained model remains at chance regardless of dataset size.

3.3.2 Train/Validation/Test Split

The dataset is partitioned 70/15/15 into training, validation and test sets by stratified random sampling, preserving class proportions across all three. A fixed random state ($\texttt{seed} = 42$) ensures all models and conditions are evaluated on identical samples.

3.3.3 Exploratory Data Analysis and Preprocessing

Preprocessing decisions were determined empirically using a `RepeatedStratifiedKFold` ($k = 5$, $n_{\text{repeats}} = 3$) protocol yielding 15 paired evaluations per pipeline, with supporting tables in Appendix B.

Features are normalised with `RobustScaler` (median subtraction, interquartile-range scaling), fitted *exclusively* on the training split to prevent leakage. It was preferred over `StandardScaler` empirically, achieving macro-F1 0.718 ± 0.004 against 0.713 ± 0.004. Where noise is injected on the Python side, perturbations are applied in raw voltage space and re-transformed with the training-fitted scaler, keeping the noise model's physical interpretation intact.

The full 208-dimensional vector is retained. Two reductions were tested and rejected: correlation pruning at $\rho = 0.75$ cost 3.5 pp macro-F1 and Principal Component Analysis (PCA) at 95.3% variance cost 4.5 pp, each ranking below the full-feature pipeline in all 15 folds. Aggressive reduction destroys the local structure the convolutional filters exploit, and Appendix B.4 shows this matters far more under noise than these clean-data margins suggest.

3.4 Model Architectures

Three classical baselines and a purpose-designed 1D-Convolutional Neural Network (CNN) are evaluated under identical data conditions, addressing the gap identified in Section 2.3.4: an SVM (Cortes and Vapnik, 1995), an RF (Breiman, 2001), and an MLP sized to match the CNN head's fully connected capacity, isolating the effect of convolutional feature extraction (full configurations in Appendix C.1).

3.4.1 1D Convolutional Neural Network

The 1D-CNN, designated `EITConv1D`, is purpose-designed for voltage difference vectors.

The architecture treats the 208-element voltage vector as a single-channel 1D signal, since adjacent measurements correspond to spatially related electrode pairs (Table 3.3). Three convolutional blocks span 2–3 electrode pairs each, batch normalisation (Ioffe and Szegedy, 2015) stabilises training, and adaptive average pooling (LeCun, Bengio,

Table 3.3: 1D-CNN architecture for EIT touch classification. "BN" denotes batch normalisation; B is the batch dimension. The adaptive average pool collapses the temporal dimension, making the head independent of input length.

Layer	Output shape	Parameters
Input	$(B, 1, 208)$	Single channel
Conv1D + BN + ReLU + MaxPool	$(B, 32, 104)$	$k = 5$, stride 1, pad 2, pool 2
Conv1D + BN + ReLU + MaxPool	$(B, 64, 52)$	$k = 5$, stride 1, pad 2, pool 2
Conv1D + BN + ReLU + MaxPool	$(B, 128, 26)$	$k = 5$, stride 1, pad 2, pool 2
AdaptiveAvgPool1d	$(B, 128, 1)$	Length-agnostic
Flatten	$(B, 128)$	—
Linear + ReLU + Dropout	$(B, 128)$	$p = 0.3$
Linear (output)	$(B, 5)$	—

and Hinton, 2015) makes the head length-agnostic. Dropout (Srivastava et al., 2014) (raised for noise-augmented training, Table 3.4) regularises the fully connected head, giving 69,189 parameters in total.

3.5 Training Protocol

3.5.1 CNN Training

The CNN is trained using the Adam optimiser (Kingma and Ba, 2015) with the hyperparameters in Table 3.4. Two safeguards, with values set experimentally, prevent overfitting: ReduceLROnPlateau and early stopping (Prechelt, 1998), always retaining the best model.

Table 3.4: CNN training hyperparameters.

Parameter	Value
Learning rate η	10^{-3}
Weight decay λ	10^{-4}
Batch size	64
Maximum epochs	200
LR reduction factor	0.5 (after 10 epochs without improvement)
Early stopping patience	40 epochs
Dropout p	0.3 (0.4 for noise-augmented training)
Label smoothing ϵ (noise-augmented)	0.05

For noise-augmented training, label smoothing (Müller, Kornblith, and Hinton, 2019) and increased dropout (Srivastava et al., 2014) add further regularisation. Online domain randomisation (Tobin et al., 2017) samples fresh noise draws for each batch, applied in raw voltage space before re-scaling into the model input, preserving the noise model's physical interpretation.

The loss function is cross-entropy over the five classes:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^{N} \sum_{c=1}^{5} y_{i,c} \log \left(\text{softmax}(\hat{y}_{i,c})\right) \tag{3.5}$$

where $y_{i,c}$ is the one-hot encoded true label and $\hat{y}_{i,c}$ are the raw logits, suiting single-label multi-class classification and penalising confident misclassification strongly.

3.5.2 Hyperparameter Selection

Baseline models are trained on the same training split using scikit-learn (Breiman, 2001; Cortes and Vapnik, 1995). The MLP uses its own internal early stopping, while the SVM and RF models are trained to convergence.

Hyperparameters for all four families are fixed *a priori* rather than tuned per condition, following canonical references for each family (Breiman, 2001; Cortes and Vapnik, 1995; LeCun, Bottou, et al., 1998). A condition-specific search would confound the noise-augmentation effect with capacity and regularisation changes, and bias comparisons toward whichever condition received the larger search budget (Caruana and Niculescu-Mizil, 2006; Henderson et al., 2018). To confirm the augmentation benefit is not a hyperparameter artefact, learning rate, dropout (Srivastava et al., 2014) and weight decay are swept separately, each retrained from scratch (Figure 3.4).

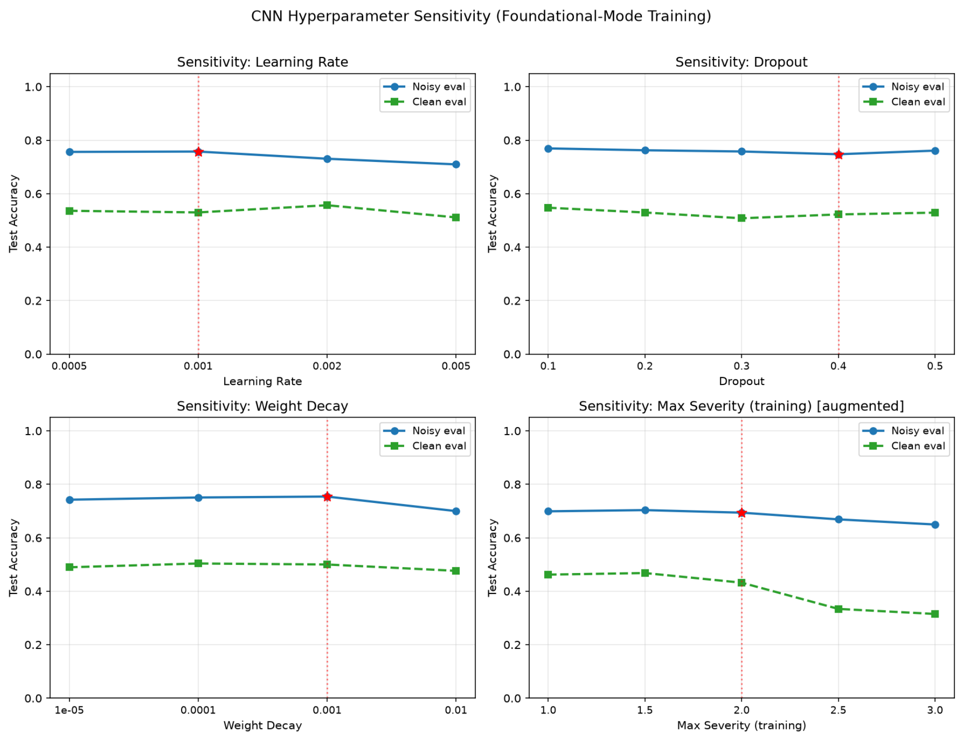


Figure 3.4: CNN hyperparameter sensitivity analysis. The retained defaults (shown by a red star) lie within the high-performing region, indicating that the main noise-augmentation result is not driven by a narrow hyperparameter choice.

3.5.3 Reproducibility

All experiments use fixed random seeds: NumPy, PyTorch and MATLAB are all seeded at 42. The complete configuration is stored in a YAML file, enabling exact reproduction from a single command.

3.6 Evaluation Framework

3.6.1 Evaluation Conditions

To quantify the effect of noise augmentation, all models are evaluated under four conditions forming a 2×2 matrix:

Table 3.5: Four-condition evaluation matrix. Each cell represents a distinct experiment.

	Evaluate on Clean	**Evaluate on Noisy**
Train on Clean	Ceiling	Vulnerability (sim-to-real gap)
Train on Noisy	Generalisation	Robustness (target condition)

Noisy$\rightarrow$Noisy is the operationally relevant cell, since every real measurement carries hardware noise, and the augmentation benefit is read as the improvement from Clean$\rightarrow$Noisy to Noisy$\rightarrow$Noisy. Domain adaptation methods (Ganin et al., 2016; Tzeng et al., 2017) are not used as comparators, since they require unlabelled target-domain samples that a simulation-only study cannot supply.

3.6.2 Performance Metrics

Two metrics are reported: overall accuracy and macro-averaged F1, which weights all classes equally and is more informative given the classes' differing signal magnitudes. Per-class precision, recall and F1 appear in confusion matrices. Headline values are computed on the held-out test split; validation metrics serve only for scheduling, early stopping and model selection.

3.6.3 Validation Protocol

Each experiment is repeated over 5 independent random seeds with stratified 70/15/15 splits (Kohavi, 1995), giving the mean $\pm$ standard deviation reported throughout. Seeds vary both the data partition and weight initialisation, so the reported spread bounds their combined effect, and models are retrained from scratch each time. A validation slice is carved from the training partition within every run, so early stopping never observes held-out data.

3.6.4 Robustness Evaluation

Following the multi-severity paradigm of Hendrycks and Dietterich (2019), trained models are evaluated at severity multipliers from $0.0\times$ to $3.0\times$ in steps of $0.5\times$, applied to all noise parameters simultaneously for each training regime. Chapter 4 extracts three quantities: degradation from training severity to $3\times$, the *crossover severity* at which randomisation overtakes fixed-noise training, and each regime's clean-domain accuracy.

3.6.5 Ablation Protocol

The ablation isolates the contribution of each noise component, and the sensitivity of the conclusions to component ordering through four complementary designs:

1. **Core mismatch conditions**: the four-condition matrix instantiated within the ablation pipeline, for direct comparison against the variants.
2. **Single-component training**: four experiments, each with only one component active.
3. **Exhaustive subset ablation**: all $2^4 - 1 = 15$ non-empty subsets, exposing interactions that marginal single-component analysis cannot capture.
4. **Component-order ablation**: all physically admissible orderings within each subset, with quantisation last (digitisation terminates the chain) and contact impedance before electrode bias (multiplicative precedes additive).

Every configuration is scored twice, since the two protocols answer different questions and, as Chapter 4 shows, give opposite answers. *Matched* evaluation tests a model on data corrupted by the same components it was trained with, measuring the difficulty of a world containing only those components. *Deployment* evaluation tests it against the full four-component corruption, measuring what omitting components costs in the field.

Only the second is operationally meaningful, since hardware exhibits all four sources simultaneously. Matched evaluation silently adjusts the test distribution to suit the model, rewarding configurations that define an easier problem rather than those that transfer. It is retained here only as a diagnostic.

Throughout, corruption is applied in *foundational mode*: noise is drawn once per training sample and held fixed across epochs, as the pre-generated dataset behaves. This prevents the training regime and the active component set from varying together. All variants inherit the same regularisation preset as the main noise-trained condition, leaving the component set as the only difference.

3.6.6 Noise Parameter Sensitivity

Each parameter was swept independently with the others held at default (Table 3.6).

Table 3.6: Values swept for each noise parameter, others held at default.

Parameter	Swept values
SNR	$\{60, 50, 40, 30, 20\}$ dB
Contact impedance σ_z	$\{5, 10, 15, 20, 25\}\%$
Electrode bias $b_{\max}$	$\{0.005, 0.01, 0.02, 0.04, 0.08\}$
ADC resolution	$\{8, 10, 12, 14, 16\}$ bits

This addresses the concern that the parameters are nominal rather than measured: if the conclusions hold across the plausible range of each, they do not depend on a particular calibration assumption.

Chapter 4

Results

Headline values are held-out test metrics reported as mean $\pm$ standard deviation over five independent seeds with stratified 70/15/15 splits.

4.1 The Simulation-to-Reality Gap

Table 4.1 reports all four classifiers across the four-condition evaluation matrix on raw 208-dimensional features.

Table 4.1: Classification accuracy (%) on raw features across the four evaluation conditions.

Model	Clean→Clean	Clean→Noisy	Noisy→Noisy	Noisy→Clean
CNN	94.00 $\pm$ 6.89	19.98 $\pm$ 0.33	**75.81 $\pm$ 0.62**	**52.23 $\pm$ 1.94**
MLP	**98.81 $\pm$ 0.69**	21.32 $\pm$ 0.34	57.25 $\pm$ 0.80	47.61 $\pm$ 0.69
RF	91.03 $\pm$ 0.30	**22.57 $\pm$ 0.18**	57.85 $\pm$ 0.64	29.91 $\pm$ 3.00
SVM	83.38 $\pm$ 0.43	20.00 $\pm$ 0.00	50.90 $\pm$ 0.72	51.66 $\pm$ 0.41

Three findings follow. First, every model achieves strong clean-domain accuracy (83–99%), confirming the forward model produces separable class representations, so the task itself is not the limiting factor. Second, *every* model collapses to approximately chance under Clean→Noisy. The collapse is architecture-independent and feature-representation-independent (Appendix B.4), identifying the gap as *distributional* rather than a deficit of model capacity: no classifier can compensate at inference for an unseen training-time shift. This is considerably more extreme than the degradations of up to 40% reported for image classifiers by Hendrycks and Dietterich (2019).

Third, noise-aware training recovers a large fraction of the loss. The CNN reaches 75.81% under Noisy→Noisy, an improvement of 55.8 percentage points (pp) over the same architecture trained on clean data.

Reconstruction shows the same collapse. Figure 4.1 inverts representative single frames back onto the mesh. Clean measurements localise every contact class distinctly, confirming physically coherent fields and genuine class separability in image space. The same frames under the full four-component corruption reconstruct to artefact, with no recoverable contact location, and the quadrant seams throughout are the per-electrode block structure of the dominant error in the image domain. Direct classification retains 75.81%, whereas reconstruction retains nothing, the practical case for bypassing the inverse problem.

The CNN outperforms the strongest baseline on noisy data by 17.9 pp (RF, 57.85%). The natural explanation is that convolutional kernels exploit the block-structured, per-electrode form of the dominant corruption (Section 4.2), and this is directly testable. Applying a fixed permutation to the 208 measurement indices, identically to training and test data and *after* noise injection, preserves information content exactly while destroying adjacency. The CNN then loses 30.2 pp, whereas the three exchangeable baselines are unaffected, the SVM to the decimal place (Table 4.2). The permuted CNN falls below both RF and MLP, so without block structure the convolutional prior is not merely uninformative but actively harmful.

Table 4.2: Accuracy (%) on noisy evaluation before and after a fixed permutation of the measurement vector, applied after noise injection. CNN values are means over 5 seeds; baselines are deterministic given the split.

Model	Original	Permuted	Change
CNN	**76.21 ± 0.52**	**45.97 ± 1.37**	**−30.24**
RF	60.03	59.12	−0.91
MLP	56.96	56.99	+0.03
SVM	52.53	52.53	0.00

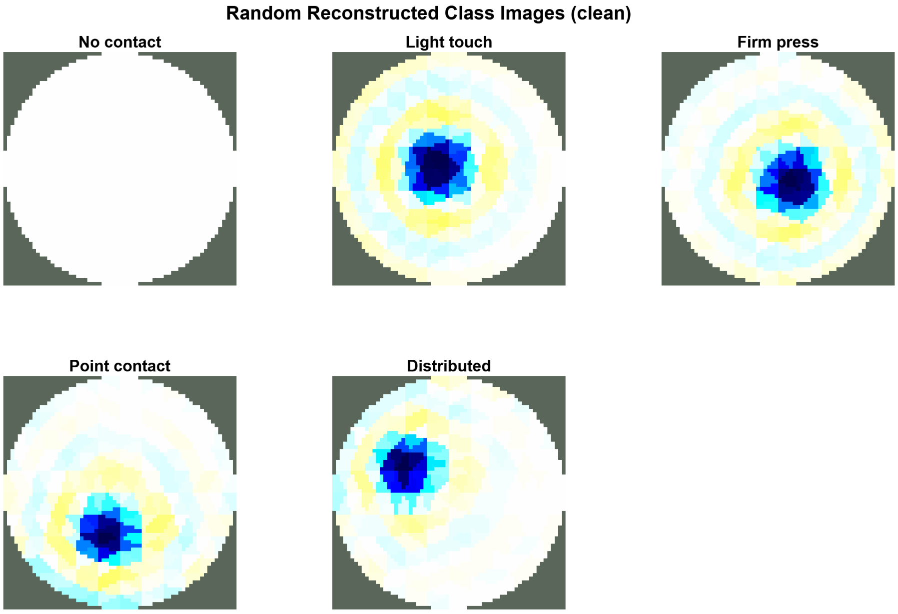
4.1.1 Training Regimes

Table 4.3: CNN accuracy (%) on noisy evaluation by training regime, raw features.

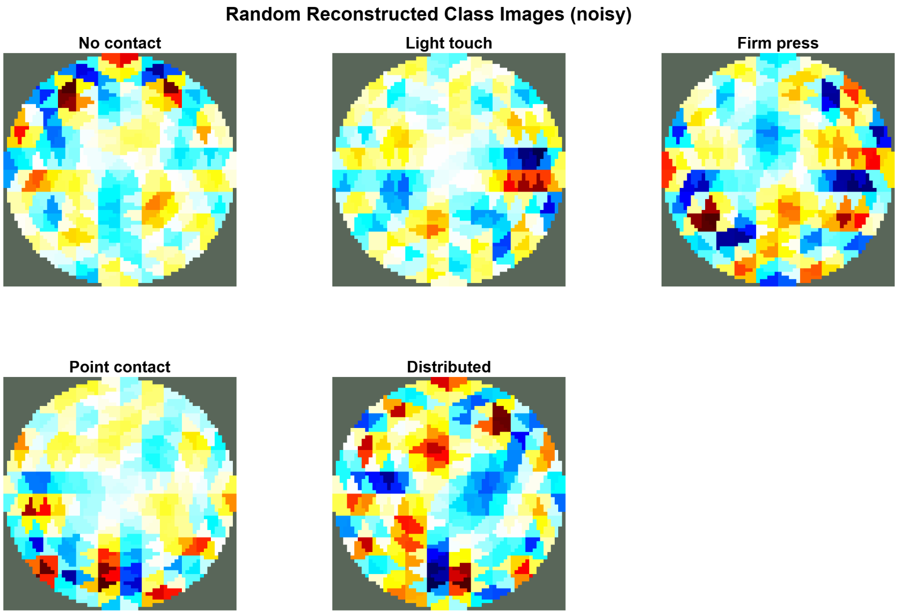
Training Regime	Accuracy
Fixed noisy dataset	**75.81 ± 0.62**
Online augmentation	69.50 ± 0.62
Mixed (30% clean + augmented)	69.17 ± 0.68

Training on the fixed, pre-generated noisy dataset gives the highest noisy-domain accuracy (75.81%), exceeding online augmentation by 6.3 pp, an order of magnitude larger than the 0.62 pp seed-to-seed spread of either condition. That advantage is specific to the training severity, however, and reverses at higher severities (Section 4.3).

Both fixed-noise and online-augmented training nonetheless lose accuracy on uncorrupted data (52.2% and 40.2%), since neither observes an uncorrupted measurement;



(a) Clean measurements.



(b) The same frames under the full four-component corruption.

Figure 4.1: Difference reconstructions of randomly drawn single frames from each class, solved with a one-step linearised Gauss–Newton method under a Laplace smoothness prior (hyper-parameter 0.003). Clean frames localise each contact class; the same frames under the four-component noise model yield no recoverable contact location, and the quadrant seams expose the per-electrode block structure of the corruption. The dissertation classifies the boundary voltages directly and never solves this inverse problem; the reconstructions are shown to characterise the measurement corruption, not as a proposed method.

the no-contact class especially seen only in perturbed form. Mixed training, retaining 30% clean samples per batch, recovers this to 71.50% for 0.3 pp on noisy data.

4.2 Noise Component Ablation

The ablation asks which noise components a training-time model must include, which requires care in how each configuration is evaluated. A model trained with only component X can be tested either on data corrupted by X alone (*matched*), or on the full four-component corruption (*deployment*). Only the latter is operationally meaningful, since real hardware exhibits all error sources simultaneously, so a model omitting a component must nonetheless face it in the field.

Table 4.4: Single-component ablation. Each CNN is trained with one noise component active and evaluated twice: on the same single component (matched) and on the full four-component corruption (deployment).

Training noise	Matched (%)	Deployment (%)
Gaussian only	96.61 $\pm$ 0.92	20.88 $\pm$ 1.34
Contact impedance only	90.11 $\pm$ 3.85	19.72 $\pm$ 0.20
Electrode bias only	78.68 $\pm$ 0.85	**57.79 $\pm$ 0.58**
Quantisation only	92.89 $\pm$ 3.53	14.03 $\pm$ 5.50
Full four-component	75.81 $\pm$ 0.62	

The two columns rank the components in opposite orders, and the matched column is misleading. Judged on matched evaluation, electrode bias appears the *most* damaging component (78.68%, lowest of the four) while Gaussian noise appears benign (96.61%). Judged on deployment evaluation, the conclusion reverses: models trained on Gaussian noise, contact impedance or quantisation alone all perform at or below chance, whereas the electrode-bias model retains 57.79%. The matched column measures the difficulty of an artificial world containing a single corruption, not the value of modelling it.

Two complementary claims follow. Electrode bias is *necessary*: every configuration omitting it collapses to chance under realistic conditions, regardless of its score in its own matched world. Quantisation-only training is the extreme case, falling below chance (14.03%) with high seed variance, indicating a model trained on deployment-negligible corruption learns a representation that is not merely uninformative but actively damaging. Electrode bias alone is nonetheless *not sufficient*, falling 18.8 pp short of the full model, so robust deployment requires the dominant structured error modelled jointly with the remaining sources.

4.2.1 Direct Characterisation of the Domain Gap

The dominance of electrode bias can be confirmed from the measurements themselves, independently of any classifier. Decomposing each vector into signal and noise contributions (Table 4.5) shows electrode bias accounting for effectively all noise energy in every class, exceeding the touch signal by a factor of 4.6 for the largest class and approximately 41 for light touch. The touch signature is a small perturbation on a much larger structured offset; that the offset is block-structured is what keeps the task tractable, and once discounted, the signal-to-residual ratio recovers to 11.3–11.9. Measurement-level decomposition and ablation therefore converge on the same component from independent directions.

Table 4.5: Signal and noise energy decomposition by class (L_2 norms, means over the dataset). Electrode bias accounts for essentially all noise energy and exceeds the touch signal itself by a factor of 4.6 to 41 depending on class.

Class	Signal	Total noise	Bias	Signal:residual
No contact	0.0000	0.1669	0.1656	—
Light touch	0.0040	0.1655	0.1655	11.3
Firm press	0.0248	0.1657	0.1657	11.5
Point contact	0.0058	0.1657	0.1657	11.3
Distributed contact	0.0359	0.1654	0.1653	11.9

4.2.2 Cancellation of Electrode Bias Under Differencing

Kolehmainen et al. (1997) and Boone and Holder (1996) report that difference imaging tolerates electrode error constant between the reference and measurement frames, becoming vulnerable only when it changes. Since the model above applies bias directly to $\Delta\mathbf{v}$, sweeping the fraction ρ of bias common to both frames tests this assumption directly (full derivation and results in Appendix A.6). The response is strongly non-linear: accuracy moves by under 2 pp across the first half of the range, and even at $\rho = 0.9$ a 14 pp deficit remains against full cancellation, which requires identical electrode positions and interface state at both acquisitions, the condition Boone and Holder (1996) identify as failing in practice through drift. The headline result is therefore insensitive to this assumption everywhere except a physically unrealisable limit.

4.2.3 The Minimal Sufficient Noise Model

Enumerating all component subsets identifies how much of the noise model is actually required. Table 4.6 reports the best subset at each size, ranked by deployment accuracy.

Table 4.6: Best component subset at each size, ranked by deployment accuracy.

n	**Best subset**	**Deployment (%)**	**Matched (%)**
1	Electrode bias	56.85 $\pm$ 1.89	77.33 $\pm$ 1.69
2	Gaussian + electrode bias	**76.53** $\pm$ **0.52**	76.73 $\pm$ 0.81
3	Gaussian + electrode bias + quantisation	76.90 $\pm$ 0.80	77.04 $\pm$ 0.37
4	All four components	75.81 $\pm$ 0.62	

A two-component model of Gaussian noise and electrode bias reaches 76.53%, indistinguishable from the best three-component subset (76.90%) and from the full four-component model. Contact impedance and quantisation contribute no measurable deployable robustness once the other two are present.

The claim is scoped to the four-component family: the other two are redundant *given* this noise model, not necessarily sufficient against the fuller error sources of Table 2.4. The necessity result for electrode bias carries no such caveat, since further sources could only strengthen it.

The electrode-bias-only configuration is trained in both enumerations, giving an incidental replication estimate: two independent five-seed runs return 57.79% and 56.85%, a 0.94 pp difference that sets the practical resolution of the ablation and is why the spread across the two-, three- and four-component models is read as agreement rather than an ordering.

Each required component has a distinct role. Electrode bias supplies the dominant structured distortion defining the domain shift (Section 4.2.1). Gaussian noise supplies the measurement noise floor, which matters disproportionately for the no-contact class: without it, no-contact samples are exact zero vectors in training but carry noise at test time, rendering the class undetectable. The per-component severity sweep isolates this: Gaussian noise at any non-zero severity raises accuracy by 20.1 pp, corresponding almost exactly to the 20% of samples in that class (Figure 4.2).

That mechanism suggests the two-component result decomposes further: since the no-contact class is an exact zero vector by construction, Gaussian noise's measured value may be confined to rescuing it rather than conferring robustness. Repeating the subset ablation on the four contact classes alone tests this (Table 4.7).

Table 4.7: Deployment accuracy (%) on the four contact classes only, chance $= 25\%$. Means over 5 seeds. Electrode bias alone matches the full model once the degenerate class is removed.

Training noise	**Deployment (%)**
Gaussian only	25.07 $\pm$ 0.13
Electrode bias only	70.53 $\pm$ 0.66
Gaussian + bias	71.21 $\pm$ 0.44
Full four-component	69.79 $\pm$ 1.50

Electrode bias alone now matches the full model (70.53% against 69.79%), adding Gaussian changing it by only 0.68 pp, within the seed spread, while Gaussian alone sits exactly at chance. The sufficiency result therefore separates into two claims: electrode bias alone suffices to *discriminate* contact types, and Gaussian noise is required only so contact can be *detected* at all, since a model without a noise floor cannot recognise a noisy vector as the no-contact state. The minimal model is one component for discrimination plus a noise floor for detection.

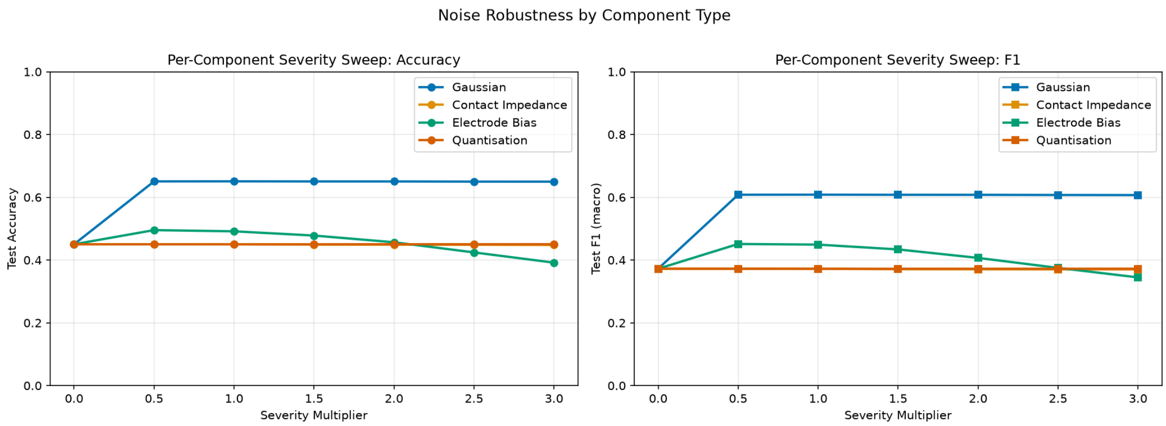


Figure 4.2: Per-component severity sweep. A CNN trained on the full noise model at 1× severity is evaluated with only one component active at varying severity. Enabling Gaussian noise recovers the no-contact class (a step of ≈20 pp, independent of severity), while electrode bias is the only component whose severity materially degrades accuracy.

4.2.4 Sensitivity to Component Ordering

Because the components are applied compositionally, their order can alter the resulting distribution. The effect is modest for the full four-component model (1.29 pp between best and worst admissible ordering) but substantial within two-component subsets: Gaussian → electrode bias achieves 76.53%, reversed order only 54.98%, a 21.6 pp difference. This is a parametrisation artefact, not a physical effect: the Gaussian component is specified relative to signal amplitude, so applying it after the much larger bias inflates its magnitude accordingly (full derivation in Appendix A.8; implications discussed in Section 5.2).

4.3 Robustness Under Severity Scaling

Each training regime was evaluated across noise severities from 0× (clean) to 3× the calibrated level.

The regimes exhibit a clear trade-off between peak accuracy and breadth of robustness (Table 4.8). Fixed-noise training is best at the severity it was trained on (75.95%

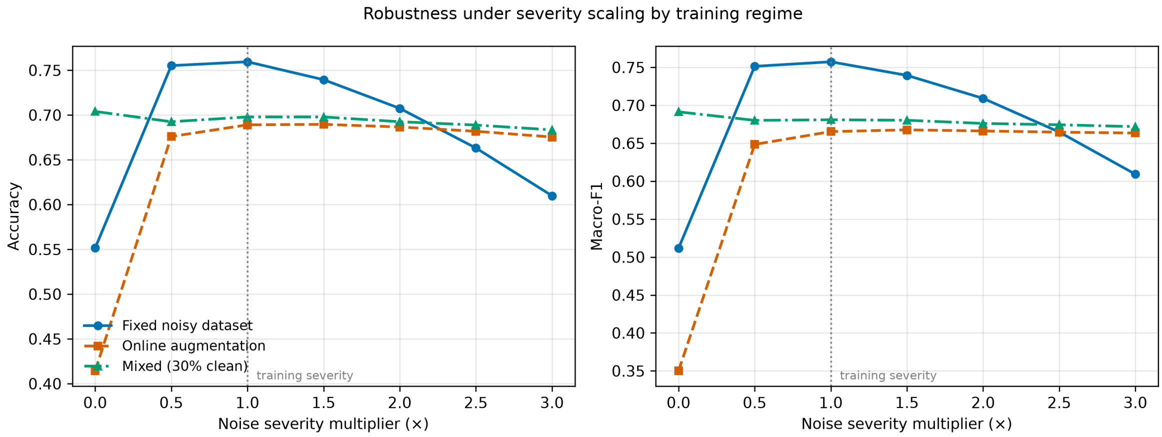


Figure 4.3: Accuracy and macro-F1 against noise severity for the three training regimes. The dotted line marks 1×, the severity at which the fixed-noise model was trained. Fixed-noise training peaks at its training severity and degrades steadily beyond it. Randomised regimes are substantially flatter and overtake it at approximately 2.2×.

Table 4.8: Accuracy against noise severity multiplier by training regime.

Regime	**0**×	**0.5**×	**1**×	**1.5**×	**2**×	**2.5**×	**3**×
Fixed noisy	55.15	75.52	**75.95**	73.95	70.75	66.32	60.99
Online augmentation	41.44	67.60	68.91	68.96	68.67	**68.19**	67.55
Mixed	**70.40**	69.25	69.79	69.79	69.25	68.88	68.35

at 1×) but degrades monotonically thereafter, losing 15.0 pp by 3×. The randomised regimes, trained across severities sampled from $[0.5, 2.0]$, are markedly flatter, with online augmentation varying by only 1.4 pp between 1× and 3×. The curves cross at approximately 2.2×, beyond which randomisation is superior.

This is the behaviour domain randomisation predicts (Tobin et al., 2017): distributing training capacity across a range of conditions costs accuracy at any single condition but buys insensitivity across the range. Mixed training additionally retains clean-domain competence, 70.40% at 0× against 41.44% for online augmentation (implications for regime selection in Section 5.3).

The 0× column is informative in itself: all three regimes lose accuracy on perfectly clean data, most severely online augmentation, largely the no-contact class (Section 4.2.3), since a model trained only on corrupted measurements has never observed a true zero vector.

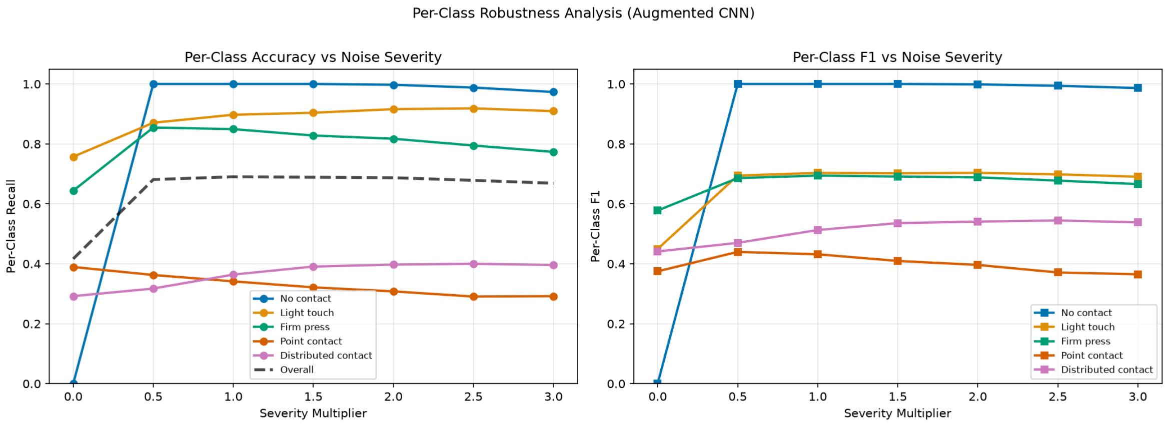


Figure 4.4: Per-class accuracy against noise severity for the noise-augmented CNN. Point contact is the weakest class throughout; the no-contact class is undetectable at 0$\times$ severity because it appears as an exact zero vector never encountered in training.

4.4 Supporting Analyses

4.4.1 Per-Class Vulnerability

Per-class behaviour (Figure 4.4) is not uniform, and the ordering is physically interpretable against the per-class signal magnitudes of Table 4.5. *Point contact* is the weakest class at every severity (39% at 0.5$\times$ falling to 29% at 3$\times$), consistent with two compounding factors: the second-smallest voltage signature ($L_2 = 0.0058$) and, as established by the mesh convergence study (Section 3.1.1), being intrinsically sub-mesh-scale, the class whose simulation is least faithful. Its poor classification is therefore attributable partly to the model and partly to the fidelity limits of its representation.

Firm press and *distributed contact*, the two largest-amplitude classes, are the most reliably recovered. *Light touch* improves slightly with severity (75.7% to 90.9%): counter-intuitive, but the same mechanism as the Gaussian result above, additional noise making the smallest-signal classes distinguishable from no-contact rather than absorbed into it. Accuracy for *no contact* is 0% at 0$\times$ and $\geq$97% at every non-zero severity.

The safety implication is specific: detecting the onset of contact, the transition most relevant to grasp control, is reliable under the trained noise conditions, but the class carrying the highest local pressure (point contact) is the least reliable, confused with other contact classes rather than no-contact. A controller could therefore be expected to detect *that* contact has occurred more reliably than to characterise *what* it was.

4.4.2 Predictive Confidence

The Bayesian tradition in EIT treats quantified uncertainty as integral to the solution (J. Kaipio and Somersalo, 2005), yet ML-based classification reports bare point predictions. Table 4.9 compares each model's mean predicted confidence against the accuracy it achieves.

Table 4.9: Mean predicted confidence against actual accuracy.

Model → Data	**Mean confidence**	**Accuracy**	**Gap**
Clean CNN → clean	96.4%	93.8%	$+2.6$
Clean CNN → noisy	99.7%	19.7%	$\mathbf{+80.0}$
Augmented CNN → clean	55.4%	39.6%	$+15.8$
Augmented CNN → noisy	64.4%	68.8%	-4.4

In its own domain, the clean-trained model is trustworthy (96.4% confidence against 93.8% accuracy), but outside it the correspondence collapses entirely (99.7% confidence at 19.7% accuracy, an 80-point gap), the most dangerous failure mode available to a safety-critical classifier since the softmax carries no warning that the model has left its domain. The noise-augmented model shows the opposite pattern on noisy data, confidence slightly *understating* accuracy (64.4% against 68.8%). Overconfidence is therefore a symptom of *domain shift* rather than an intrinsic architectural property, refining the observation of Guo et al. (2017) that modern networks are overconfident by construction.

4.4.3 Sensitivity to Modelling Assumptions

Because the noise parameters are nominal rather than measured (Section 3.2.3), each was swept independently with the others held at default (Table 4.10).

Table 4.10: CNN accuracy (%) as each noise parameter is varied independently, others held at default. Default settings in bold.

SNR (dB)	60	50	**40**	30	20
Accuracy	69.3	69.3	**69.4**	69.3	69.3
Contact imp. (%)	5	**10**	15	20	25
Accuracy	69.2	**69.4**	69.5	69.4	69.1
Bias $b_{\max}$	0.005	0.01	**0.02**	0.04	0.08
Accuracy	68.6	69.9	**69.4**	64.9	35.0
ADC (bits)	8	10	12	14	**16**
Accuracy	30.1	69.3	69.7	69.4	**69.4**

The conclusions are insensitive to SNR and to contact impedance across their full plausible ranges (variation $<$0.5 pp), reassuring since contact impedance is the parameter least well-constrained by the literature. Two parameters do matter: accuracy collapses to 35.0% when electrode bias is quadrupled to $b_{\max} = 0.08$, and to 30.1% at 8-bit ADC resolution. Both are interpretable: the former pushes the structured offset beyond the amplitude at which the model learned to discount it, while the latter raises the quantisation step until it is comparable to the touch signal itself, only 0.0040 for light touch. The engineering implications are discussed in Section 5.5.

Hyperparameter sensitivity is comparatively modest, accuracy varying by 2.2–5.4 pp across learning rate, dropout, weight decay and training severity range (Appendix A.9).

4.4.4 Generalisation Across Noise Realisations

Because train and test noise are drawn from the same parametric family, it must be confirmed that the model accommodates the noise *distribution* rather than memorising the realisations in its training set. Evaluated on ten independently generated noisy test sets, each with entirely different electrode-bias, contact-impedance and Gaussian draws, accuracy is stable at 75.61% $\pm$ 0.17%, direct evidence of distributional accommodation rather than memorisation. Whether training noise is fixed once per sample or redrawn every epoch likewise makes no measurable difference (69.09% against 69.50% for online augmentation). What matters is distributional coverage, not realisation-level correspondence: a far less restrictive requirement than device-specific calibration. Full figures are in Appendix A.5.

Chapter 5

Discussion

5.1 The Structure of the Simulation-to-Reality Gap

The simulation-to-reality gap in EIT tactile classification is dominated by a single, highly structured error source, changing what an effective response looks like. The universal collapse to chance under Clean→Noisy evaluation (≈20% across four architectures and three feature representations) shows the gap is not a capacity problem. Adding parameters, changing inductive bias or reducing dimensionality all fail, because none addresses the cause: the training distribution does not contain the deployment distribution. This extends the corruption-robustness results of Hendrycks and Dietterich (2019) to a considerably more extreme regime, consistent with the bound of Ben-David et al. (2010) in which target error is limited by source error plus distributional divergence.

The ablation locates that divergence in electrode positioning bias, independently confirmed by the energy decomposition to exceed the touch signal in every class (Section 4.2.1). A clean-trained classifier is therefore not asked to tolerate a perturbed version of a familiar input; it is presented with a vector whose dominant component it has never seen, where the informative part is a small residual, so chance-level performance is the expected outcome.

That dominant component is *structured*, constant within each electrode's measurement block and therefore confined to a low-dimensional subspace. This is what makes the problem tractable at all, and the permutation ablation confirms it is what the CNN exploits: destroying adjacency while preserving information costs 30.2 pp and leaves the exchangeable baselines untouched. The architectural result is therefore not a general claim about deep learning but a specific match between inductive bias and a known distortion geometry, which is only available to a designer who knows that geometry in advance.

This also reframes the failure of dimensionality reduction (Appendix B.4): PCA and Linear Discriminant Analysis (LDA) fitted on clean data discard the directions along which the bias acts, since those carry no clean-domain class information, optimal for the source domain, pathological for the target.

5.2 What the Ablation Implies for Noise-Model Design

Evaluating a component-ablated model on data containing only that component measures the difficulty of an artificial world, not the value of modelling the component. The two evaluations disagree completely: matched evaluation ranks electrode bias as the most damaging component and Gaussian noise as benign, while deployment evaluation shows Gaussian-only, contact-impedance-only and quantisation-only models to be worthless in the field (14.03–20.88%) whereas the electrode-bias model retains 57.79%. Had only the matched column been reported, as is the convention in the studies reviewed in Section 2.3.4, the guidance derived from it would have been exactly inverted. This correction applies to component ablations generally, not only to EIT.

Gaussian noise combined with electrode bias reaches 76.53%, on a level with the three- and four-component models. Excluding the no-contact class separates the two roles cleanly: electrode bias alone then matches the full model, so Gaussian noise contributes nothing to discriminating contact *types* and is required solely to make the no-contact state detectable. What presents as a two-component requirement is really one component for discrimination plus a noise floor for detection, and a practitioner whose device never needs to report the absence of contact would not need the second.

Contact impedance and quantisation, both physically real and carefully parametrised, contribute nothing measurable at 16-bit resolution. For contact impedance, this is less surprising than it appears: Boone and Holder (1996) found difference imaging largely insensitive to skin impedance that stays constant between the reference and measurement frames, becoming vulnerable only when it drifts, and the present model holds one factor per electrode fixed within the frame, reproducing exactly that benign case. The null result therefore confirms an established property of difference imaging rather than showing interface impedance unimportant, implying the component would matter under sequential acquisition. The corollary matters more: omitting electrode bias reduces the model to chance regardless of what else is included.

The 21.6 pp difference between the two admissible orderings of Gaussian noise and electrode bias is a parametrisation artefact rather than a physical effect: the Gaussian component is specified relative to signal amplitude, so applying it after a much larger bias inflates its magnitude accordingly. It is nonetheless a real trap for compositional noise models, invisible unless orderings are enumerated: any component defined relative to the running signal must be applied at its physically correct point in the chain.

5.3 Choosing a Training Regime

No training regime dominates, so the guidance is conditional. Where deployment noise is known and stable, fixed-noise training is preferable, since the randomisation penalty of roughly 6 pp is unjustified. Where it is uncertain or varies between devices, the realistic case for a prosthetic worn daily, randomisation trades that 6 pp against a 15.0 pp loss at elevated severities. Mixed training is the most defensible default when the environment is uncharacterised, retaining clean-domain competence while matching online augmentation on noisy data.

A secondary result is that *how* the noise is presented during training does not matter, provided the distribution is right. Noise resampled every epoch and noise fixed once per sample yield accuracies differing by less than the seed-to-seed spread (69.50% and 69.09%), and accuracy is stable across ten independent realisations (75.61% $\pm$ 0.17%). What matters is that the training distribution *covers* the deployment distribution, not that it reproduces a specific device's signature, so per-device calibration is unnecessary.

The clean-domain cost of noise-aware training is easy to overlook: every noise-trained regime loses accuracy on uncorrupted data, concentrated almost entirely in the no-contact class, safety-relevant for a device operated immediately after calibration. Mixed training mitigates it.

5.4 Validity of a Simulation-Only Study

Any classifier trained purely on synthetic data must address whether its conclusions survive real measurements. The relevant framework is the *approximation error* formulation of J. Kaipio and Somersalo (2005), relating accurate and simplified forward operators as

$$\mathbf{v}_{\text{measured}} = \mathcal{F}_{\text{accurate}}(\sigma) + \varepsilon_{\text{noise}} \approx \mathcal{F}_{\text{approximate}}(\sigma) + \varepsilon_{\text{approx}} + \varepsilon_{\text{noise}}, \tag{5.1}$$

where $\varepsilon_{\text{approx}}$ is the systematic discrepancy introduced by modelling simplifications. Nissinen, Heikkinen, and J.P. Kaipio (2008) showed that characterising this term statistically and folding it into the noise model substantially improves EIT reconstruction on real hardware.

The classical inverse crime (J. Kaipio and Somersalo, 2007; Wirgin, 2004) arises when the same discretisation generates the data and solves the inverse problem, so that discretisation errors cancel. The present work does not solve an inverse problem, so there is no mechanism for numerical cancellation. The concern does not vanish, since a classifier trained on one discretisation could still learn features specific to it, but the cross-mesh transfer matrix shows a model trained on the medium mesh loses only 4.4 pp on meshes it never saw (Section 3.1.1). More tellingly, the dominant learned behaviour, discounting a block-structured electrode offset, is a property of the electrode model rather than the interior discretisation, and mesh-specific exploitation would not survive both independent noise realisations *and* three discretisations.

Viewed through Equation 5.1, the electrode-level components model exactly the interface idealisation the Complete Electrode Model introduces, so electrode bias dominating the gap is evidence that the dominant approximation error here is the electrode interface, not mesh discretisation, consistent with Vilhunen et al. (2002). Where the Bayesian approach characterises $\varepsilon_{\text{approx}}$ statistically and folds it into the likelihood, noise-augmented classification achieves a comparable effect by enriching the training distribution.

5.5 Deployment Readiness

The parameter sweep (Section 4.4.3) shows the conclusions are insensitive to SNR across 60–20 dB and to contact-impedance variation across 5–25%, which matters since contact impedance is the least well-constrained parameter in the model. Accuracy falls sharply at quadrupled electrode bias and at 8-bit ADC resolution, so these two bound the regime in which the findings hold. Read as engineering guidance, this would prioritise electrode placement tolerance and ADC bit depth over front-end noise performance, though the sweep varies one parameter at a time about a single operating point, making it a robustness check rather than a specification.

Under domain shift, the clean-trained model reports 99.7% mean confidence while achieving 19.7% accuracy, and no downstream logic can recover a signal the softmax does not carry. That a distribution-matched model restores the correspondence does not license complacency, since the deployment distribution is never known exactly. Quantified uncertainty is integral to the Bayesian EIT tradition (J. Kaipio and Somersalo, 2005), and ML-based classification's abandonment of it for bare point predictions is a regression that should be reversed before any deployment claim.

At approximately 76% five-class accuracy under matched noise, the system is not deployment-ready, and that figure is an upper bound for a simulation-only study, not a projection for hardware. The per-class analysis is more informative than the aggregate: detecting that contact occurred is reliable, characterising its type is not, so a staged deployment using contact detection for control while treating finer discrimination as advisory would be consistent with the evidence.

These conclusions complement rather than compete with hardware advances. Morphology optimisation (Dong et al., 2025) improves the signal at the electrodes; noise-aware training addresses the uncertainty that remains. Since the dominant gap sits at the electrode interface rather than the substrate, the two address different bottlenecks and should compose.

5.6 Ethical, Economic and Sustainability Implications

The confidence result carries the sharpest ethical consequence: a prosthetic that fails *silently* endangers its user in a way that one failing loudly does not, and a model reporting 99.7% certainty at chance accuracy fails in precisely that manner. The safety-relevant quantity is therefore not aggregate accuracy but the correspondence between confidence and correctness, together with the per-class breakdown showing which failures are averaged away. Reporting both is an obligation, not a refinement, since the aggregate figure alone would have concealed the failure entirely.

Economically, a simulation-first approach lowers the barrier to EIT research, using only freely available software on standard hardware, which matters since limb loss falls disproportionately on low- and middle-income populations (McDonald et al., 2021), where high-cost sensing risks widening the inequality it aims to relieve. Establishing which noise components matter before a device is built transfers design cost from fabrication to computation, the cheaper half for those groups.

That shift also carries the sustainability argument: dataset generation takes 10–20 minutes and training under an hour on a consumer GPU, so the direct environmental cost is negligible against the physical prototyping it displaces, and halving the noise model while showing per-device calibration unnecessary both remove costs that would otherwise scale with production.

5.7 Limitations

The following limitations bound the generalisability of these findings.

1. **No hardware validation.** All results are simulation-based, and the noise model is literature-justified rather than calibrated against a specific device. Since electrode bias is the decisive component, an error in its distributional form would propagate directly to the central conclusions, making this the most important item of future work.

2. **Structure of electrode bias.** Bias is drawn independently per electrode and uniformly distributed. The inter-frame part of that assumption is now measured (Section 4.2.2), but *spatial* correlation between neighbouring electrodes, which a hydrogel deforming as a sheet under strain would produce, is not. Correlated bias, occupying a lower-dimensional subspace, should be easier for a convolutional model to discount, so the reported figures may be pessimistic here. Measuring the true bias covariance on a physical sensor remains the natural first hardware experiment.

3. **Incompleteness of the noise model.** Four error sources are modelled; several documented others are not (Table 2.4), making the reported gap a conservative

lower bound, since further sources could only worsen transfer, though it confines the sufficiency claim to the modelled family. Whether two components remain sufficient once drift and geometric deformation are included is untested.

4. **2D geometry.** The forward model uses a 2D cross-section, omitting out-of-plane current flow and inter-ring electrode coupling, a deterministic discrepancy that noise augmentation cannot compensate for (Section 5.4).

5. **Static single-frame classification.** Temporal correlations are not exploited, and drift, documented over hours in hydrogel substrates, cannot be represented within a single measurement frame.

These follow from a simulation-first design chosen to isolate the effect of noise augmentation under controlled conditions. They constrain the confidence with which specific accuracy figures can be projected onto hardware, but leave the principal findings intact, since those are comparative and structural rather than absolute.

Chapter 6

Conclusions

6.1 Summary of Findings

This dissertation asked whether physically motivated, multi-component noise augmentation can bridge the simulation-to-reality gap for EIT tactile classification, and which components such a model must contain.

1. **The gap is severe, universal and distributional.** All four classifiers, across all three feature representations, collapse to chance under realistic multi-component noise after clean-data training. Independent of architecture and representation, the collapse cannot be addressed by model capacity.

2. **Noise-matched training substantially closes it.** A 1D-CNN reaches 75.81% against 19.98% for the clean-trained equivalent, a gain of 55.8 pp.

3. **Electrode positioning bias dominates the gap**, accounting for effectively all noise energy, exceeding the light-touch signal roughly 41-fold. Component ablation, severity response and direct energy decomposition converge on this independently.

4. **Modelling electrode bias is necessary; two components suffice, and they play separate roles.** Every noise model omitting electrode bias performs at or below chance, while Gaussian noise plus electrode bias reaches 76.53%, level with the full four-component model. Excluding the degenerate no-contact class separates the roles: electrode bias alone suffices to discriminate contact types; Gaussian noise is required only to make contact detectable.

5. **The architectural advantage is caused by block structure, and the gap survives partial cancellation.** A fixed permutation of the measurement vector, information-preserving but adjacency-destroying, costs the CNN 30.2 pp while leaving the exchangeable baselines unchanged. Sweeping the fraction of electrode bias com-

mon to both frames leaves accuracy within 2 pp up to 50% cancellation, with a 14 pp deficit at 90%.

6. **The ablation evaluation protocol changes the conclusion.** Matched evaluation ranks the components in the opposite order to deployment evaluation, making Gaussian noise appear benign and electrode bias worst, so reporting matched results alone inverts the guidance they yield.

7. **Training regime is a characterised trade-off.** Fixed-noise training maximises accuracy at its training severity (75.95% at $1\times$) but loses 15.0 pp by $3\times$, whereas randomised training is far flatter, overtaking it at approximately $2.2\times$.

6.2 Achievement of Objectives

Objective 1 was met: the noise model was developed and validated by parameter sensitivity analysis, but determining which components a training model must actually contain required first correcting the evaluation protocol, since the originally planned matched-condition design would have produced the opposite conclusion, itself among the more transferable outcomes of the work. Objective 2 was met as specified; the dataset was additionally validated by a mesh convergence study. Objective 3 established the CNN's advantage under noise, now traced to block structure by permutation, and Objective 4 was extended into a severity trade-off analysis identifying a crossover between regimes.

Of the contributions claimed in Section 1.5, the structured per-electrode noise model, minimal sufficient model and ablation-protocol correction are all established above, with the sensitivity analysis bounding the regime in which they hold.

6.3 Recommendations and Future Work

Two findings can be adopted immediately at no cost: component ablations should be evaluated against the full corruption a deployed device meets rather than the reduced corruption trained on, which applies generally and not only to EIT, and Gaussian-only augmentation should no longer be accepted as evidence of robustness, with multi-component noise including electrode bias as the minimum standard.

Two design choices follow: the training regime should be selected by how well the deployment environment is characterised rather than by peak accuracy at any one severity, fixed noise where the level is known and stable, randomisation where uncertain, and mixed training by default. In hardware, electrode placement tolerance and ADC resolution ($\geq$12 bits here) warrant explicit budgets, while front-end SNR and interface impedance uniformity can be relaxed, though that ordering needs confirming on a physical device first.

That qualification identifies the priority: validation against real measurements. Every conclusion about electrode bias rests on an assumed uniform, independent per-electrode distribution, so measuring the true distribution on a fabricated 16-electrode skin would either confirm these results or propagate an error through all of them. Until then, the accuracies reported here remain matched-distribution figures rather than demonstrated sim-to-real performance.

Three further extensions address model fidelity: **three-dimensional forward modelling** would remove the largest deterministic approximation error, which randomisation cannot absorb; **temporal modelling** over sequential frames would admit drift compensation and represent the slow-varying sources excluded here by construction (Table 2.4); and **morphology-aware randomisation** would test whether the minimal sufficient model survives a change of sensor design (Dong et al., 2025).

6.4 Reflections on the Research Process

Two methodological lessons emerged, each from an error corrected during the work rather than from the original plan.

The first concerns evaluation protocol. The ablation was initially designed to train and test each configuration on the same noise components, the intuitive reading of "isolating a component", and produced a coherent but entirely inverted account of which components matter. In transfer settings, the evaluation distribution encodes a claim about the deployment context, so choosing it casually is a modelling error, not a presentational detail.

The second concerns internally consistent but unverified results. Several analyses were at one stage produced by a pipeline whose reference model had been trained under the wrong regime, so the outputs were plausible, self-consistent and wrong, caught only because the severity sweep disagreed with the main results table for what should have been an identical measurement. Cross-checking analyses that ought to agree proved far more effective than scrutinising any single result, and that check would be automated if the project were repeated.

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Appendices

Appendix A

Supplementary Results

This appendix provides supplementary figures and tables supporting the methodology of Chapter 3 and the findings of Chapter 4: mesh selection (Section A.1), training convergence, full confusion matrices, corroboration of the validation protocol by an independent cross-validation (Section A.4), generalisation across noise realisations (Section A.5), the bias-cancellation derivation (Section A.6), the inadequacy of Gaussian-only augmentation (Section A.7), detailed ablation results, predictive-confidence diagnostics and the sensitivity analyses.

A.1 Mesh Selection

Three matched noisy datasets were generated at coarse (`c`), medium (`d`) and fine (`f`) refinement with identical class definitions, noise model, splits and hyperparameters; one CNN was trained per mesh and evaluated on all three (Table A.1). Training on `c` gives the highest in-mesh score but the largest cross-mesh variation (8.35 pp), whereas `d` and `f` transfer more stably (4.37 and 3.47 pp). Mesh `d` is retained as it balances literature comparability against stable cross-mesh behaviour, and since complexity and runtime are comparable (Table A.2), the decision rests on generalisation rather than cost (Section 3.1.1).

A.2 Training Convergence

Figures A.1–A.3 show training and validation loss curves for the CNN under the key training regimes. All three converge stably, confirming that the accuracy differences reported in Chapter 4 reflect what each regime learns rather than any failure to optimise. The noise-trained models converge more slowly than the clean-trained model, consistent with the more complex decision boundaries required when a large structured offset must be discounted.

Table A.1: Cross-mesh CNN performance (%) under matched training protocol. Rows denote training mesh, columns denote test mesh.

Train mesh	Test c	Test d	Test f	Mean	Range
c (Accuracy)	83.92	75.57	77.09	78.86	8.35
d (Accuracy)	79.44	75.07	75.76	76.76	4.37
f (Accuracy)	78.77	75.31	76.88	76.99	3.47
c (Macro-F1)	84.02	74.88	76.86	78.59	9.14
d (Macro-F1)	79.52	74.90	75.67	76.57	4.62
f (Macro-F1)	78.33	74.13	76.06	76.17	4.20

Table A.2: Computational profile by training mesh.

Train mesh	Train time (s)	Inference (ms/sample)	Parameters	Peak memory (MB)
c	277.15	0.0149	69,189	61.45
d	275.00	0.0104	69,189	36.90
f	275.28	0.0105	69,189	36.90

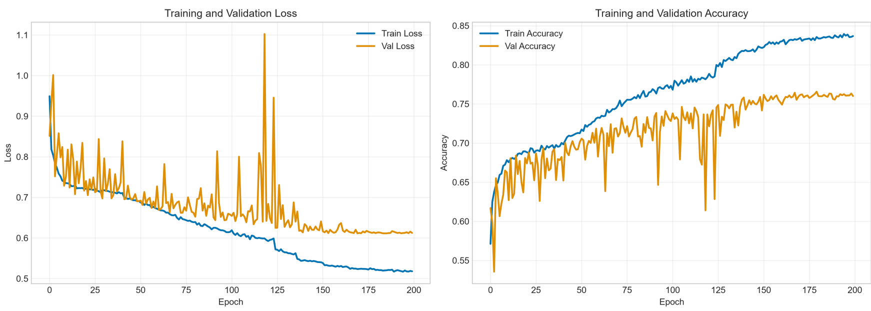

Figure A.1: CNN training curves: fixed noisy training data, evaluated on noisy test data (Noisy→Noisy condition). The model converges within approximately 60 epochs with smooth validation loss reduction.

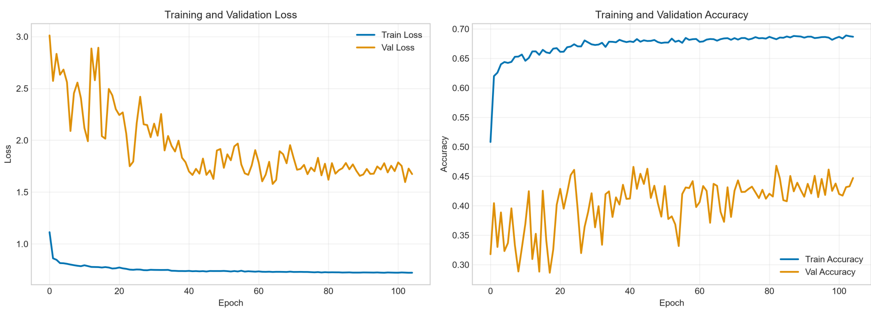

Figure A.2: CNN training curves: online noise augmentation, evaluated on noisy test data (Augmented→Noisy condition). Convergence is stable, but to a lower plateau than fixed-noise training, reflecting the additional variability the model must accommodate when noise is resampled each epoch.

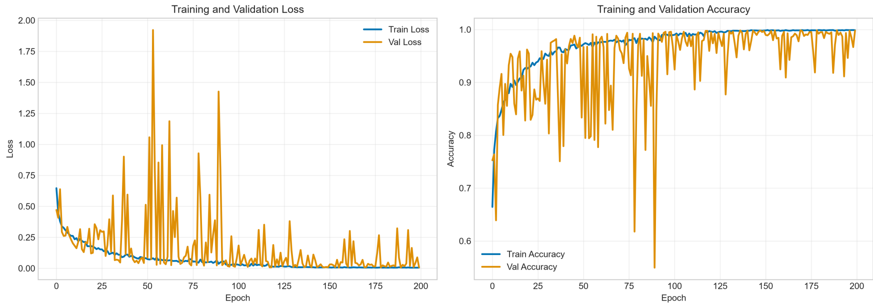

Figure A.3: CNN training curves: clean training data, evaluated on clean test data (Clean→Clean condition). Rapid convergence confirms the forward model produces well-separated class representations.

A.3 Confusion Matrices

Figures A.4–A.6 present the full confusion matrices for the CNN under key evaluation conditions.

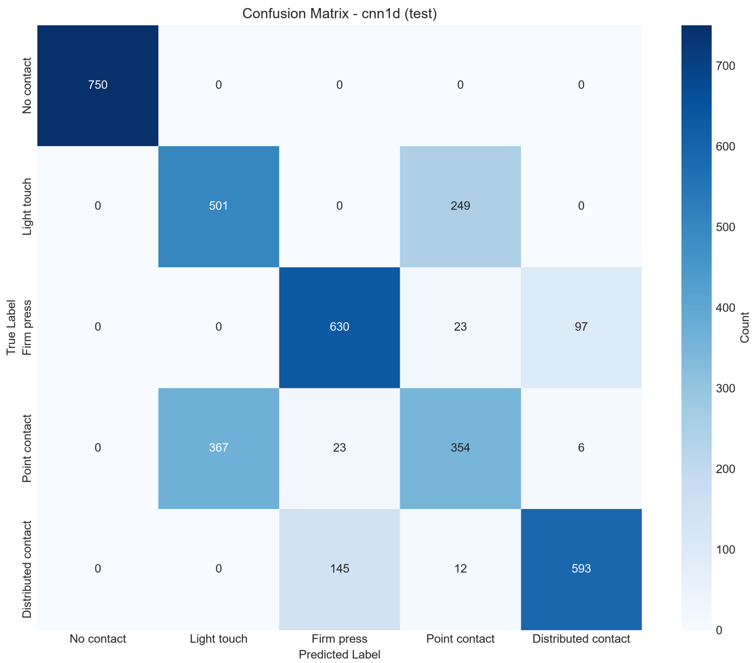

Figure A.4: Confusion matrix: CNN trained on noisy data, evaluated on noisy data (Noisy→Noisy, 75.8%). Principal confusions involve point contact, the class with the smallest spatial footprint and the least faithful simulation (Section 4.4.1).

A.4 Corroboration by Cross-Validation

All headline results in Chapter 4 are reported over five independent seeds with stratified 70/15/15 splits. As an independent check on that protocol, a stratified 5-fold cross-validation (Kohavi, 1995) was also run, in which each fold's test partition is disjoint from

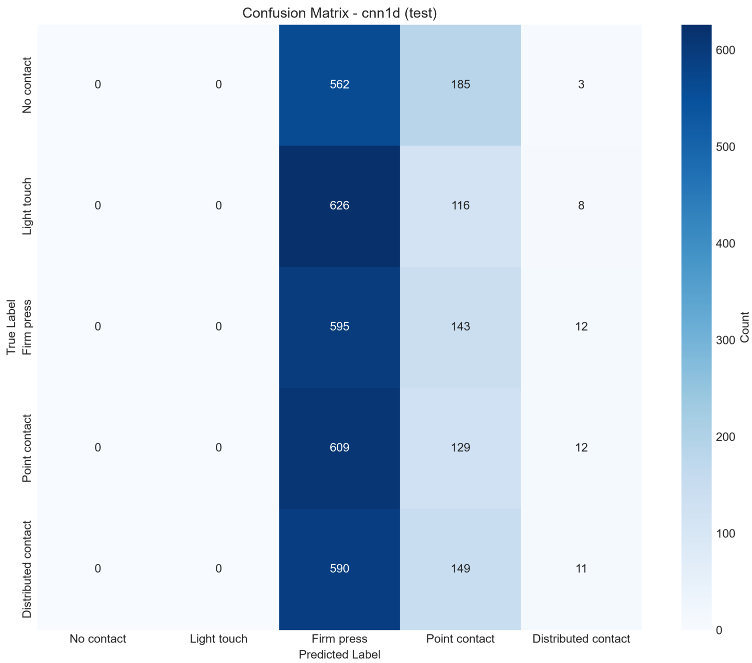

Figure A.5: Confusion matrix: CNN trained on clean data, evaluated on noisy data (Clean→Noisy, 20.0%). Predictions collapse primarily onto a single class, indicating complete loss of discriminative structure rather than uniform random guessing.

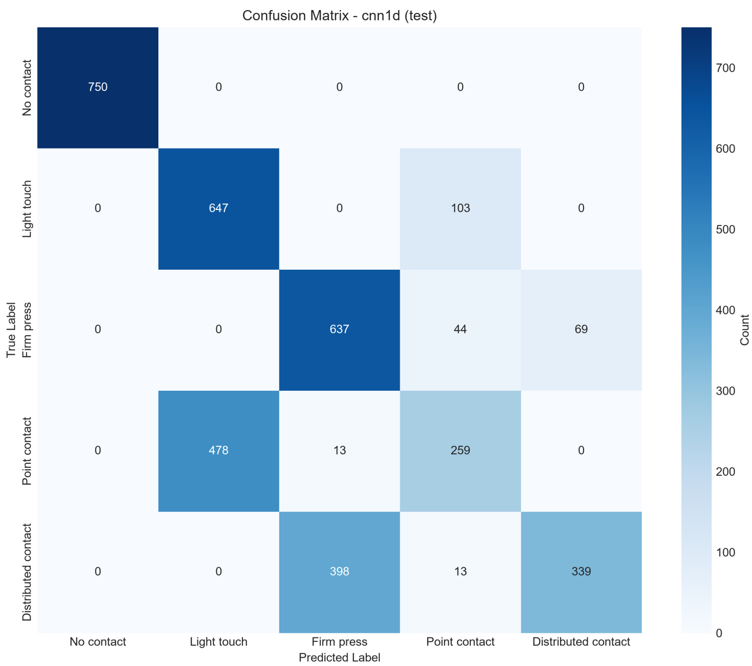

Figure A.6: Confusion matrix: CNN with online noise augmentation, evaluated on noisy data (Augmented→Noisy, 69.5%). The error structure closely resembles that of fixed-noise training (Figure A.4), indicating that the two regimes learn qualitatively similar representations and differ mainly in the sharpness of the resulting decision boundaries.

every other, and every sample is held out exactly once; within each fold a validation slice is carved from the training partition, so early stopping never observes held-out data, and models are retrained from scratch per fold. Table A.3 compares the two schemes.

Table A.3: CNN accuracy (%) under the multi-seed protocol used throughout Chapter 4 against an independent stratified 5-fold cross-validation. The two schemes vary different things: folds vary the data partition, and seeds vary the partition and initialisation together.

Condition	**5 seeds**	**5-fold CV**
Clean→Clean	94.00 ± 6.89	96.26 ± 1.89
Clean→Noisy	19.98 ± 0.33	19.66 ± 0.79
Noisy→Noisy	75.81 ± 0.62	75.58 ± 0.76
Noisy→Clean	52.23 ± 1.94	52.52 ± 1.31
Augmented→Noisy	69.50 ± 0.62	69.16 ± 0.25

The two schemes agree closely on every condition, which establishes that the reported differences reflect the training condition rather than either source of sampling variability. The one divergence is Clean→Clean, where cross-validation gives 96.26% ± 1.89 against 94.00% ± 6.89 across seeds. That gap is itself informative, isolating the CNN's clean-domain instability as an *initialisation* effect rather than a data-partition one. The larger multi-seed spread is the more conservative estimate and is the figure quoted in Table 4.1.

Every difference discussed in Chapter 4 exceeds the seed-to-seed standard deviation by a wide margin: the 55.8 pp recovery from noise-aware training is roughly ninety times the 0.62 pp spread of the noisy-trained condition, and the narrowest comparison examined, the 6.3 pp between fixed-noise and online augmentation, is ten times it.

A.5 Generalisation Across Noise Realisations

Supplementary detail for Section 4.4.4. The trained noisy-domain CNN was evaluated on ten independently generated noisy test sets, each with entirely different electrode-bias vectors, contact-impedance factors and Gaussian realisations. Accuracy was 75.61% ± 0.17% across the ten draws, against 74.37% on the original MATLAB-generated test set. The variation across realisations is far smaller than any effect discussed in Chapter 4. That the alternative draws score marginally *higher* reflects the small implementation difference between the MATLAB and Python noise paths, and bounds it at approximately 1.2 pp.

A CNN was also trained with noise sampled once per training sample and held fixed across all epochs, mimicking a population of devices each with its own persistent electrode characteristics. It achieved 69.09% on noisy evaluation and 69.47% on an independent draw, a 0.41 pp difference from online augmentation (69.50%) smaller than the seed-to-seed spread of either condition. Whether training noise persists per sample or is redrawn each epoch therefore has no measurable effect on deployed accuracy.

A.6 Cancellation of Electrode Bias Under Differencing

Supplementary derivation and results for Section 4.2.2. Writing the per-electrode offset as a shared component plus independent parts, the shared term cancels exactly under differencing and the residual acting on $\Delta\mathbf{v}$ scales as $(1-\rho)$, where ρ is the fraction common to both frames. Table A.4 sweeps ρ to test this directly.

Table A.4: CNN accuracy (%) against ρ, the fraction of electrode bias common to the reference and measurement frames and therefore cancelled by differencing. Means over 5 seeds. $\rho = 0$ is the model used throughout; $\rho = 1$ is static misplacement.

ρ	**0.00**	**0.25**	**0.50**	**0.75**	**0.90**	**1.00**
Accuracy	75.80	76.46	77.47	78.53	82.85	**96.99**

A.7 Inadequacy of the Gaussian-Only Noise Model

Table A.5 evaluates the noise model used almost universally in prior EIT work, sweeping Gaussian noise alone across a 40 dB range of SNR.

Table A.5: CNN and Random Forest accuracy (%) under Gaussian-only noise at varying SNR. Accuracy is essentially invariant across the full range for every model tested.

Model	**60 dB**	**50 dB**	**40 dB**	**30 dB**	**20 dB**
CNN (clean-trained)	79.7	79.7	79.7	79.7	79.6
CNN (Gaussian-augmented)	72.5	72.6	72.5	72.7	72.8
CNN (full-noise-augmented)	78.9	78.9	78.9	78.7	78.8
Random Forest (Gaussian-trained)	91.0	91.1	90.8	90.8	90.7

Accuracy varies by under 0.5 pp across a 40 dB range, four orders of magnitude in noise power, for every model tested. A clean-trained classifier that collapses to 19.98% under the full four-component noise model is essentially unaffected by Gaussian corruption even at 20 dB, since additive zero-mean noise is largely absorbed by batch normalisation (Ioffe and Szegedy, 2015) and does not disturb the geometry separating the classes.

This governs how the accuracies reported in the reviewed literature should be read. Robustness demonstrated under Gaussian-only augmentation (H. Chen et al., 2023; Park, Yuk, et al., 2022b) is not evidence of robustness to the distortions that dominate real measurements, and is the basis for the recommendation in Section 6.3 that multi-component noise, including electrode biases, be treated as the minimum standard for such a claim. The result is consistent with the ablation of Section 4.2, where a Gaussian-only training model reaches only 20.88% under deployment-realistic evaluation.

A.8 Ablation Details

Figures A.7–A.9 supplement Section 4.2. They are reported under *matched* evaluation and are included principally to document the protocol whose conclusions Chapter 4 shows to be misleading; the deployment-realistic results that supersede them are in Tables 4.4 and 4.6.

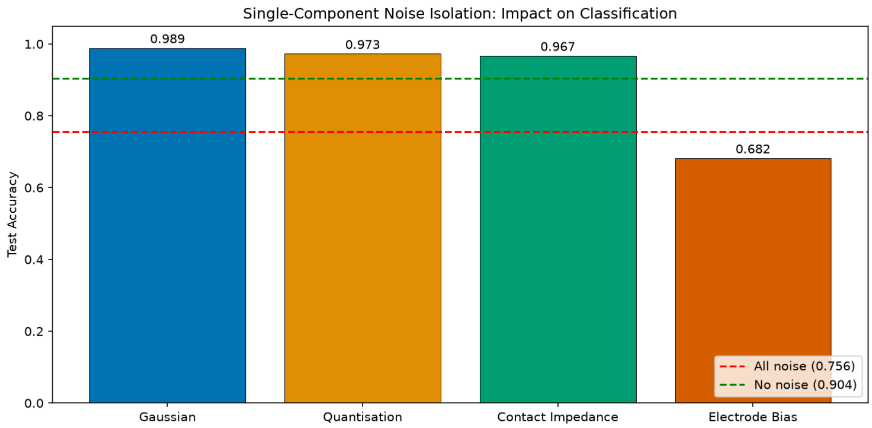

Figure A.7: Single-component ablation under *matched* evaluation, in which each model is tested on the same single component it was trained with. Read in isolation, this figure is misleading: it ranks electrode bias worst (78.68%) and Gaussian noise best (96.61%), the reverse of the deployment-realistic ranking in Table 4.4. It is retained to make that discrepancy explicit.

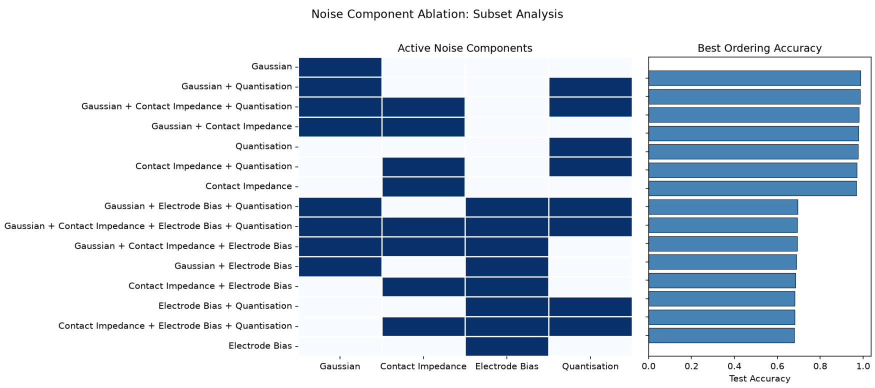

Figure A.8: Matched-condition accuracy across all 15 non-empty component subsets. Subsets containing the electrode bias cluster within 75.3–77.8%, while those omitting it reach 85.3–97.2% under matched evaluation, higher precisely because they define an easier world rather than better training distributions. Under deployment evaluation, the two groups separate in the opposite direction, at 54.7–76.9% and 16.5–23.4% respectively (Section 4.2).

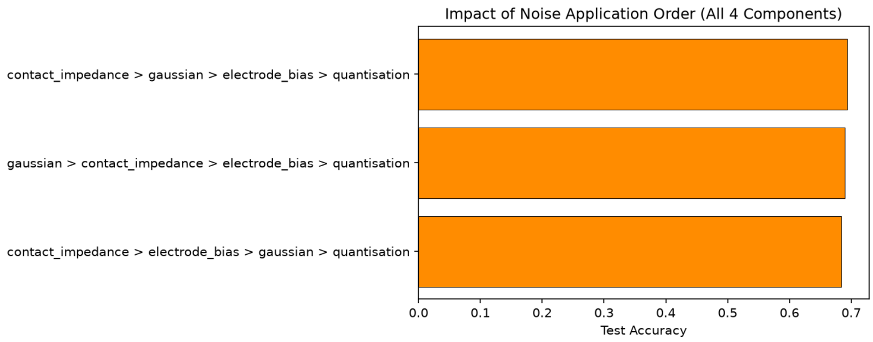

Figure A.9: Effect of component application order for the full four-component model, where the spread is modest (1.29 pp between the best and worst admissible permutation). Ordering matters far more within two-component subsets, though only under deployment evaluation: reversing Gaussian and electrode bias costs 0.67 pp matched but 21.6 pp deployed (Section 4.2.4). The cause is that the Gaussian component is specified relative to signal amplitude (Equation 3.1): applied after the bias rather than before, its scale is set by signal-plus-bias, roughly forty times larger for light touch (Section 4.2.1), injecting far more Gaussian energy and matching deployment less closely.

A.9 Sensitivity Analyses

Figures A.10 and A.11 supplement Section 4.4.3, covering sensitivity to the noise-model parameters and to the training hyperparameters, respectively.

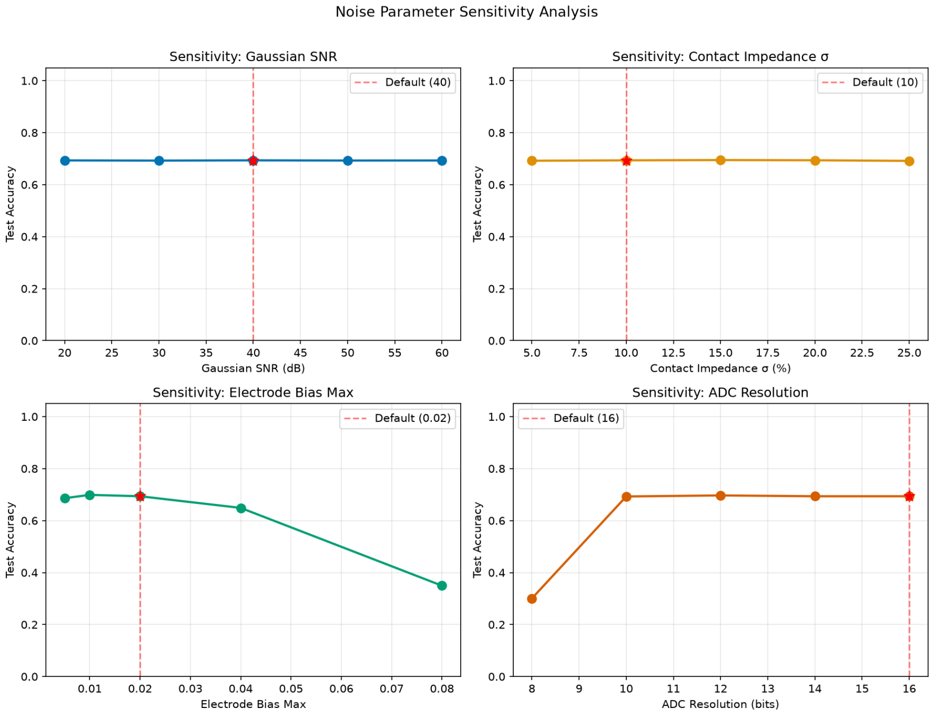

Figure A.10: Noise parameter sensitivity: CNN accuracy as each noise parameter is swept independently while the others are held at default. Accuracy is flat across the full plausible range of SNR and contact impedance, but collapses at quadrupled electrode bias ($b_{\max} = 0.08$: 35.0%) and at 8-bit ADC resolution (30.1%). See Table 4.10.

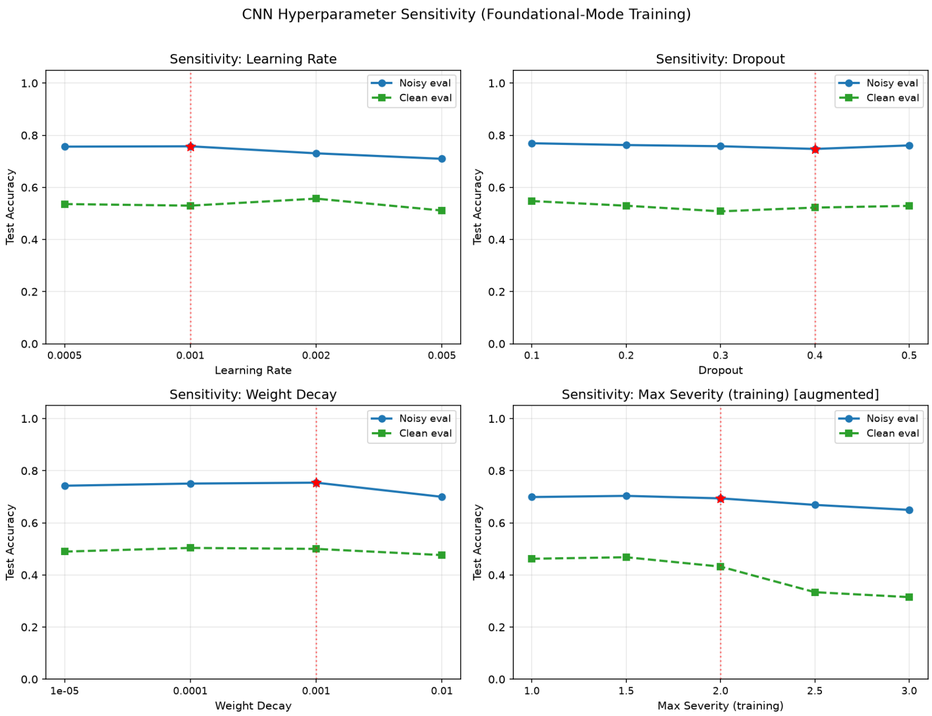

Figure A.11: Hyperparameter sensitivity: CNN accuracy under sweeps of learning rate, dropout, weight decay and training severity range. Accuracy varies by 2.2–5.4 pp across all four sweeps, confirming that the reported findings are not an artefact of the specific training configuration.

Appendix B

Exploratory Data Analysis

This appendix reports the empirical evidence underpinning the preprocessing decisions documented in Section 3.3.3: choice of feature scaler, decision to retain the full 208-dimensional voltage vector, and decision not to deduplicate the no-contact class. Section B.4 then revisits the dimensionality-reduction decision under noise, where its consequences prove far larger than the clean-data margins suggest. Except where stated, results were produced by the notebook `notebooks/eda_analysis.ipynb`.

B.1 Comparison Protocol

Seven candidate preprocessing pipelines were compared using a logistic-regression probe under a `RepeatedStratifiedKFold` cross-validation design ($k = 5$, $n_{\text{repeats}} = 3$, fixed seed 42), yielding 15 paired evaluations per pipeline. The probe is deliberately simple: its purpose is to provide a fast, model-agnostic surface on which to compare *preprocessing* effects without confounding from architecture tuning. The leakage-safe protocol fits the scaler and any reduction transform on the training partition of each fold and applies the fitted transform to the held-out partition.

B.2 Pipeline Comparison

Table B.1 reports the cross-validated macro-F1 and balanced accuracy for each candidate pipeline.

Although the absolute differences in Table B.1 are small, the ranking is consistent fold by fold: `robust_lr` achieves the highest macro-F1 in every one of the 15 evaluations, and both correlation pruning and PCA rank below the full-feature pipelines in all of them. That consistency, rather than the size of the gap, is the basis for the selection. The ranking was further confirmed on a stratified held-out partition disjoint from the cross-validation folds, which reproduced the same ordering, so `RobustScaler` with the full

Table B.1: Cross-validated comparison of preprocessing pipelines (15 paired folds, logistic-regression probe). `robust_lr` = `RobustScaler` + full 208 features; `standard_lr` = `StandardScaler` + full features; `corr`ρ = correlation-based feature pruning at threshold ρ; `pca95` = PCA retaining 95% variance (7 components); `lda_classifier` = LDA-transformed features (4 components).

Pipeline	Macro-F1 (mean±std)	Bal. Acc.	Notes
`robust_lr`	$\mathbf{0.718 \pm 0.004}$	$\mathbf{0.721}$	Selected baseline.
`standard_lr`	0.713 ± 0.004	0.717	-0.5 pp vs. robust.
`corr090_f_lr`	0.704 ± 0.004	0.709	Prune at $\rho = 0.90$.
`lda_classifier`	0.683 ± 0.110	0.700	High variance across folds.
`corr075_f_lr`	0.683 ± 0.007	0.688	Prune at $\rho = 0.75$.
`pca95_lr`	0.673 ± 0.005	0.678	7 components retain 95.3% variance.
`corr075_pca95_lr`	0.672 ± 0.007	0.677	Combined reduction.

feature set is adopted as the default preprocessing path for all subsequent experiments in Chapter 4.

B.3 Duplicate Audit

An exact-row duplicate audit on the full 25,000-sample dataset identified 4,999 duplicates within the no-contact class (Class 0) and zero duplicates across all four contact classes. The class-0 duplicates are a physically correct consequence of the homogeneous-baseline definition: every no-contact sample produces an identical conductivity image, and, in the clean (pre-noise) representation, an identical voltage difference vector of $\mathbf{0}$. Deduplication is therefore not applied. After noise injection, the class-0 samples become distinguishable through the Gaussian noise floor of Component 1 (Section 3.2.2), so no clustering pathology arises during training.

B.4 Effect of Feature Reduction

Dimensionality reduction was evaluated as an alternative to the full 208-dimensional representation. Table B.2 reports the best model per condition across the three feature spaces.

On clean data, both reductions remain competitive: LDA retains 91.43% with only four dimensions, confirming that the class-discriminative structure is genuinely low-dimensional. Under noise they are actively harmful, halving Noisy→Noisy accuracy from 75.81% to approximately 38%. Because the projections are estimated from the covariance of *clean*

Table B.2: Best accuracy (%) per condition across feature representations. Both reductions are fitted on clean training data.

Condition	Raw (208-D)	PCA (7-D)	LDA (4-D)
Clean→Clean	98.81 (MLP)	93.56 (MLP)	91.43 (RF)
Clean→Noisy	22.57 (RF)	24.43 (RF)	20.10 (RF)
Noisy→Noisy	75.81 (CNN)	38.48 (SVM)	38.14 (RF)
Noisy→Clean	52.23 (CNN)	45.42 (MLP)	22.35 (RF)

data, they retain only the directions that separate classes in the absence of noise; the dominant corruption acts along different directions (Section 4.2.1), so the information a model would need to discount it is discarded before training begins. This is a concrete instance of the result of Ben-David et al. (2010) that minimising source-domain error is insufficient when the target distribution differs.

Appendix C

Implementation Details

C.1 Baseline Classifier Configurations

Settings for the three classical baselines of Section 3.4. All are fixed *a priori* from canonical practice for each model family and held constant across every experimental condition, so that the estimated effect of noise augmentation is not confounded with per-condition tuning effort.

Table C.1: Baseline classifier configurations.

Model	Configuration
SVM	RBF kernel, $C = 10$, $\gamma =$ scale; one-versus-one decomposition (10 pairwise classifiers aggregated by voting), as implemented by `scikit-learn`'s `SVC`. The high C reflects well-separated classes in the normalised feature space (Cortes and Vapnik, 1995). Probability estimates obtained via `CalibratedClassifierCV`.
RF	500 trees, unlimited depth, minimum 5 samples per split and 2 per leaf, Gini impurity, bootstrap aggregation. The large ensemble stabilises predictions (Breiman, 2001).
MLP	208→128→128→5, ReLU activation, Adam ($\alpha = 10^{-3}$), L_2 regularisation ($\lambda = 10^{-4}$), batch size 64, early stopping on a 10% internal validation split, maximum 500 iterations.

C.2 Code Repository

The full source code, trained models, and data generation pipeline are available at:

`https://github.com/ronniepiku/eit-sim2real`

The repository includes:

- MATLAB scripts for EIDORS-based dataset generation (`matlab/`)
- Python training and evaluation pipeline (`src/eit_sim2real/`)

- Configuration files for exact reproduction (`src/eit_sim2real/configs/config.yaml`, `matlab/configs/noise_params.yaml`)
- All trained model weights and generated results (`results/`)
- Jupyter notebook for exploratory data analysis (`notebooks/`)
- Automated test suite (`tests/`)